



Research paper

The hidden cost of an image: Quantifying the energy consumption of Artificial Intelligence image generation

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ABSTRACT

With the growing adoption of Artificial Intelligence (AI) image generation, in conjunction with the ever-increasing environmental resources demanded by AI, we are urged to answer a fundamental question: What is the environmental impact hidden behind each image we generate? In this research, we present a comprehensive empirical experiment designed to assess the energy consumption of AI image generation through diffusion models. Our experiment compares 17 state-of-the-art image generative models by considering multiple factors that could affect their energy consumption, such as model quantization, image resolution, and prompt length. Additionally, we consider established image quality metrics to study potential trade-offs between energy consumption and generated image quality. Results show that image generation models vary drastically in terms of the energy they consume, with up to a 46x difference. Image resolution affects energy consumption inconsistently: increasing resolution fourfold (from 512× 512 to 1024× 1024) yields energy increases ranging from 1.3x to 4.8x across models. U-Net-based models tend to consume less than Transformer-based one. Model quantization instead results to deteriorate the energy efficiency of most models, while prompt length and content have no statistically significant impact. Results show that, albeit not common, an effective trade-off between image generation energy consumption and its quality is possible, with few of the models producing the highest quality images also being among the most energy efficient ones.

1. Introduction

Text-to-image models have seen rapid adoption in the most recent years and are, with high probability, here to stay. Albeit the numerous benefits image generation models offer, a looming question remains to date unanswered: *What is the environmental impact of image generation models?* The ever-increasing environmental impact of Artificial Intelligence (AI) is becoming an established concern in academic literature that can no longer be neglected (Verdecchia et al., 2023). Although past studies covered a wide range of topics related to the environmental sustainability of AI, to the best of our knowledge, no research effort so far considered the energy required by image generation models.

In this research, we aim to gain a comprehensive understanding of the energy consumption of image generation models by presenting a comprehensive empirical experiment comprising 17 distinct state-

of-the-art diffusion models. The experiment is conducted considering various aspects that may influence model energy efficiency, such as model quantization, image resolution, and prompt length. To gain systematic and sound insights, our research method comprises multiple experimental re-runs and statistical analyses, leading to the execution of more than 9k distinct experimental measurements. In addition to model energy consumption, we complement our results by evaluating also the quality of the generated images, achieved via the use of consolidated metrics specifically designed to measure the quality of generated images.

On one hand, our research targets researchers interested in understanding and improving the environmental sustainability of image generation, by providing a comprehensive analysis on the factors influencing energy consumption of this process. On the other hand,

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our study is intended to reach a wider audience, by providing concrete evidence grounded in empirical data of the to date overlooked environmental cost hidden behind image generation.

1.1. Related work

Recent research has highlighted the substantial carbon footprint and high energy consumption associated with training large deep learning models. However, relatively few studies focus on their efficiency during inference, despite its critical role in real-world deployment.

Gowda et al. (2023) analyze the energy consumption of various deep learning models, providing a detailed trade-off between accuracy and efficiency. Their study quantifies the energy consumption for both training and testing in terms of backbone architecture, dataset, GFLOPs (Giga Floating Point Operations Per Second), and parameter count, by disregarding however image generation tasks.

A large-scale assessment on the environmental impact of machine learning is conducted by (Castaño et al., 2023). They evaluated the carbon footprint of 1417 models hosted on Hugging Face, covering diverse domains such as multimodal models, audio, and computer vision. Notably, they find no statistically significant differences in carbon footprint between pre-trained and fine-tuned models.

As other pioneering study in this space, Budenny et al. (2022) assesses the CO₂ emissions of text-to-image generative models. They focus only on two models, Malevich and Kandinsky (Razhigaev et al., 2023), evaluating their consumption based on training epochs, loss, GPU/CPU usage, and batch size.

Among the models considered in this work, only PixArt- α (Chen et al., 2023) reported CO₂ emissions. We therefore assert that, while commonly computational cost of training is considered – typically in terms of GPU hours or Neural Function Evaluations (NFEs) – the environmental impact of models is most commonly overlooked.

Synthetic image generation models are usually evaluated using both quantitative metrics, such as FID (Fréchet Inception Distance) (Heusel et al., 2017), Precision–Recall (Sajjadi et al., 2018), IS (Inception Score) (Salimans et al., 2016), and CLIPScore (Contrastive Language-Image Pre-training Score) (Hessel et al., 2021), as well as qualitative user studies. FID and Precision–Recall require a reference dataset, representing the real data distribution against which distances are measured. In contrast, IS operates without the need of a reference dataset. All these discussed metrics rely on features extracted by a pre-trained InceptionV3 (Szegedy et al., 2016) network. Unlike them, CLIPScore does not assess image quality directly but instead measures how well the generated images align with the given prompts.

Some research has explored the impact of image resolution on generation quality. PixArt- σ (Chen et al., 2024) finds that increasing resolution improves both FID and CLIP scores, testing images at 512 and 1024 pixels with aspect ratios between 1 and 9. Similarly, Lumina (Gao et al., 2024) provides qualitative comparisons of images upscaled from 512 to 2048 pixels, benchmarking against PixArt- α (Chen et al., 2023). Stable Diffusion 1.5 (Rombach et al., 2022) examines 4× upscaling with FID and IS metrics.

Beyond resolution, studies investigate how architectural choices influence generative performance. Factors such as number of diffusion steps, guidance scale, and network type (e.g., UNet or LoRA) are frequently considered. Stable Diffusion XL Turbo (Sauer et al., 2025) explores variations in loss functions, discriminator types, conditioning methods (image- or text-based), and initialization strategies. Stable Diffusion 3 (Esser et al., 2024) studies the impact of model depth and guidance weight on FID, while Flash Stable Diffusion (Chadebec et al., 2024) and Flash Stable Diffusion XL (Chadebec et al., 2024) analyze the effects of timestamp sampling.

By considering studies focusing on the energy consumption of AI models inference at large, previous work focused on heterogeneous aspects, such as AI deployment strategies (Li et al., 2025; Husom et al., 2025; Jiang et al., 2025), load balancing strategies (Li et al.,

2023; Jin and Yang, 2025; Xie and Fang, 2025), dynamic model selection (Cruciani and Verdecchia, 2025; Nijkamp et al., 2024; Matathammal et al., 2025), clustering techniques (Stojkovic et al., 2025; Hisaharo et al., 2024; Wilkins et al., 2024), and code generation energy efficiency (Solovyeva and Castor, 2026; Wei et al., 2023; Ashraf et al., 2025).

Albeit the vast plethora of literature on Green AI inference, to the best of our knowledge, no prior work has systematically examined the energy consumption of image generation by analyzing the interplay between model architecture, quantization, resolution, and prompt length. In this study, we provide new insights into the environmental impact of diffusion models during inference, while also investigating the relationship between a model’s environmental footprint and the quality of the generated images.

2. Preliminaries

In this section, we provide an overview of the key concepts related to image generation with diffusion models, that are at the basis of this study. Specifically, we focus on two main classes of diffusion models: **U-Net-based models**, that leverage U-Net backbone, and **Transformer-based models**, that replace the U-Net with a Vision Transformer.

2.1. U-net-based diffusion models

Diffusion models have emerged as a leading paradigm in image generation. Originally introduced by Sohl-Dickstein et al. (2015), Denoising Diffusion Probabilistic Models (DDPMs) (Ho et al., 2020) involve a forward diffusion process and a reverse denoising process. These early models are defined as *unconditional*, as they generate images without external input.

In the *forward diffusion process*, Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ is progressively added to an input image $x_0 \sim q(x_0)$, sampled from the data distribution, over T timesteps, following a noise schedule. By timestep T , the image is transformed into pure noise. This process is modeled as a Markov chain, defined by $q(x_1, \dots, x_T) := \prod_{t=1}^T q(x_t | x_{t-1})$ where $q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$. In this formulation, t and β_t denote the timestep and noise schedule, respectively.

On the other side, the *reverse denoising process* consists of removing the noise and reconstructing the original image x_0 , starting from the completely noisy version, $x_T \sim \mathcal{N}(0, I)$. This is done by iteratively denoising each timestep, using a U-Net based neural network predicting the mean and the variance of the denoised image at each timestep: $p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$, where $\mu_\theta(x_t, t)$ and $\Sigma_\theta(x_t, t)$ represent the mean and variance of the Gaussian distribution.

To improve the efficiency of diffusion model training on limited computational resources without compromising quality and flexibility, Rombach et al. (2022) introduced latent diffusion models (LDMs). LDMs use a Variational Autoencoder (VAE) to encode the high-resolution input image into a low-dimensional latent representation, allowing the U-Net to perform the diffusion process with reduced computational complexity. Additionally, a CLIP-based text encoder (Radford et al., 2021) converts input text prompts into embeddings that condition the U-Net and effectively guide the diffusion process. These kind of models are categorized as *conditional*.

2.2. Transformer-based diffusion models

Transformer-based diffusion models (Peebles and Xie, 2023) replace the commonly-used U-Net backbone with a transformer (Vaswani, 2017) that operates on latent patches. Starting from Gaussian noise, a transformer network reverses the diffusion process to generate the target image, leveraging diffusion timesteps to generate distinct features at each stage.

These models support class conditioning, enabling to generate images that correspond to specific class labels. This feature allows the network to produce images from predefined categories by conditioning the transformer on both the class label and the timestep embedding, making it highly flexible for controlled image generation.

The architecture of these models follows the Vision Transformers (ViT) (Alexey, 2020), combining multi-head self-attention (MHSA) and a multi-layer perceptron (MLP) blocks, described as $X \leftarrow X + \alpha \text{MHSA}(\gamma X + \beta)$, $X \leftarrow X + \alpha' \text{MLP}(\gamma' X + \beta')$, where $X \in \mathbb{R}^{N \times C}$ represents image tokens, with N as the number of tokens and C as the channel dimension. The parameters $\alpha, \gamma, \beta, \alpha', \gamma', \beta'$ come from adaptive layer normalization (adaLN) integrating class condition embedding E_{cls} and timestep embedding E_t .

3. Methodology

This section documents the methodology used to study the energy consumption of different diffusion models during the image generation process. To ensure in-depth analysis, a wide set of diffusion models was selected. The study investigated how changes of key factors, such as models, quantization, image resolution, and prompt length, impact energy consumption. In addition to the energy consumption analysis, we also conducted a quality assessment of the generated images to explore the relationship between model energy consumption and output quality. The details of our experimental setup are presented in the following sections.

3.1. Experimental variables

To evaluate the energy consumption of different diffusion models during the image generation process, we selected a diverse set of diffusion models, including both U-Net-based and Transformer-based ones. These categories were based on their different backbone architecture, resulting in distinct image generation processes and therefore potentially different energy consumption patterns.

In designing our experiments, we identified key variables to assess their influence on energy consumption while ensuring the fairest comparison possible between models. To analyze model behavior across multiple realistic usage scenarios, we selected **model quantization** (i.e., a technique to reduce computational and memory costs by representing weights and activations with low-precision data types), **image resolution** (i.e., the resolution of the generated image), and **prompt length** (i.e., the length in number of words of the input prompt used for generating the image) in as the primary experimental variables. Conversely, we excluded the number of diffusion steps, as each diffusion model is typically designed to operate with a specific step count, which can range from as few as 2 steps to 50 or more. Running a model designed for 2 steps with 50 steps, or *vice versa*, would not only degrade image quality but also introduce significant biases in the comparison, as models are not intended to function under such conditions.

3.1.1. Diffusion models selection

For this study, we selected a total of **17 diffusion models**, based on their underlying architectures: U-Net-based and Transformer-based models. By including models from both groups, we aimed to capture a comprehensive range of diffusion-based generation techniques.

In selecting the models, we prioritized state-of-the-art models over older ones, with a focus on the models that are more widely used. We also included a few slightly outdated models that remain popular and continue to be highly downloaded by end users. In this way, we ensured a representation of both cutting-edge technologies and well-established ones for our experimentation. Furthermore, we selected models that are free for academic use and accessible offline, i.e., excluding subscription-based solutions such as DALL-E (Ramesh et al., 2022) and Midjourney (Midjourney, 2022), which are not publicly accessible and hence we could not experiment on. All chosen models

are available on Hugging Face, allowing for their standardized use and minimizing implementation variability. The models selected for our experimentation are:

U-Net-based: Stable Diffusion 1.5 (SD_1.5) (Rombach et al., 2022), Stable Diffusion XL (SDXL) (Podell et al., 2023), Stable Diffusion XL Turbo (SDXL_Turbo) (Sauer et al., 2025), Stable Diffusion XL Lightning (SDXL_Lightning) (Lin et al., 2024), Hyper Stable Diffusion (Hyper_SD) (Ren et al., 2024), Segmind Stable Diffusion 1B (SSD_1B) (Gupta et al., 2024), Latent Consistency Model Segmind Stable Diffusion 1B (LCM_SSD_1B) (Luo et al., 2023), Latent Consistency Model Stable Diffusion XL (LCM_SDXL) (Luo et al., 2023), Flash Stable Diffusion (Flash_SD) (Chadebec et al., 2024), Flash Stable Diffusion XL (Flash_SDXL) (Chadebec et al., 2024).

Transformer-based: PixArt- α (PixArt_Alpha) (Chen et al., 2023), PixArt- σ (PixArt_Sigma) (Chen et al., 2024), Flash PixArt (Flash_PixArt) (Chadebec et al., 2024), Stable Diffusion 3 (SD_3) (Esser et al., 2024), Flash Stable Diffusion 3 (Flash_SD3) (Chadebec et al., 2024), Lumina-Next-SFT (Lumina) (Gao et al., 2024), Flux.1 schnell (Flux_1) (Black Forest Labs, 2023).

Many of the models considered in this study are derivatives of a base model that has been distilled or otherwise modified to enhance image quality and improve efficiency. For instance, Stable Diffusion XL serves as the foundation for multiple variants, including Turbo, Lightning, LCM, and Flash. While these versions all originate from the same base model, they integrate distinct optimizations and refinements that influence the image generation process, potentially leading to energy consumption variations.

Table 1 reports, for each model, the recommended default configuration, including the suggested image resolution and number of inference steps, as specified in the respective model cards. The number of inference steps for each model was fixed to the default value for which the model was designed across all the experiments.

3.1.2. Model quantization

To address the substantial memory demands of diffusion models, quantization emerged as an effective technique for reducing memory usage while preserving high model performance. Given the widespread use of quantization, we study the energy consumption of each investigated model in **both quantized and unquantized forms**. For the unquantized configurations, we followed the default precision settings specified in the respective model cards. Specifically, all models were loaded in FP16, except for FLUX.1, which was loaded in BF16, and Flash-SD, Flash-SDXL, and Lumina, which were loaded in FP32.

For the quantized configurations, we utilized Quanto,¹ a quantization toolkit built on PyTorch provided through Hugging Face Optimum, a suite of tools for hardware optimization. Among the available quantization options, we selected **int8 quantization**, as it is a widely used approach that significantly reduces memory usage while maintaining acceptable image quality. Specifically, we quantized the U-Net and Transformer components using the `quantize` function provided by the Optimum Quanto library with 8-bit integer weights, while keeping all other parameters at their default values. We acknowledge that this choice represents only one of the possible quantization strategies, and that the energy consumption measured for the Quanto-based int8 quantization may not generalize to other quantization schemes.

3.1.3. Image resolution

Image resolution is a critical factor that influences both computational complexity and visual quality of generated images. Higher resolutions generally produce higher visual quality, while being characterized by increased computational requirements, and hence higher memory consumption, processing times, and potentially energy consumption.

¹ <https://github.com/huggingface/optimum-quanto> (version 0.2.6).

Table 1

Default inference model settings, including the suggested number of diffusion steps and the corresponding input image resolution, as specified in the Hugging Face model cards.

Type	Model	Default steps	Suggested resolution
U-Net-based	SD_1.5 (SD-v1.5, 2026)	50	512 × 512
	SDXL (SDXL, 2026)	50	1024 × 1024
	SDXL_Turbo (SDXL0Turbo, 2026)	1	512 × 512
	SDXL_Lightning (SDXL-Lightning, 2026)	4	1024 × 1024
	Hyper_SD (Hyper-SD, 2026)	1	1024 × 1024
	SSD_1B (SSD-1B, 2026)	50	1024 × 1024
	LCM_SSD_1B (LCM SSD-1B, 2026)	4	1024 × 1024
	LCM_SDXL (LCM SDXL, 2026)	4	1024 × 1024
	Flash_SD (FlashSD, 2026)	4	512 × 512
	Flash_SDXL (FlashSDXL, 2026)	4	1024 × 1024
Transformer-based	PixArt_Alpha (PixArt-XL, 2026)	20	1024 × 1024
	PixArt_Sigma (PixArt-Sigma-XL, 2026)	20	1024 × 1024
	Flash_PixArt (FlashPixart, 2026)	4	1024 × 1024
	SD_3 (SD-3 Medium, 2026)	28	1024 × 1024
	Flash_SD3 (FlashSD3, 2026)	4	1024 × 1024
	Lumina (Lumina, 2026)	30	1024 × 1024
	Flux_1 (FLUX 1-schnell, 2026)	4	1024 × 1024

In our study, we considered two widely used square resolutions, 512×512 and 1024×1024 , along with a non-square resolution, namely 768×1024 . This approach allowed us to explore the potential impact of aspect ratio variations on generation performance and visual quality of the images.

3.1.4. Prompt length

Understanding whether longer prompts introduce unexpected computational overhead or efficiency variations in the image generation process is important for optimizing prompt engineering strategies for different applications. Subtle variations in processing time, memory usage, or energy consumption could arise when handling more complex or descriptive input prompts.

To evaluate the effect of prompt length on energy consumption, we considered three different lengths: short, medium, and long. For **short-length prompts**, we randomly selected 30 classes from CIFAR-100 (Krizhevsky et al., 2009) dataset and we used them without any adaptation as short prompts. Therefore, short prompts consist of one or two words as presented in the original dataset. The CIFAR-100 prompts were then expanded into **long-length prompts** by using the Phi-3-Mini-4 K-Instruct (Abdin et al., 2024) large language model (LLM), a lightweight and state-of-the-art open model, specifically designed for general-purpose AI systems. We used 13 examples from Stable Diffusion's best-performing photographic prompts (Wind, 2023) as a reference, instructing the LLM to add additional details to the input short prompt. Therefore, long prompts consists of an average of 33 words. Finally, long prompts were manually truncated to obtain **medium-length prompts**, consisting of an average of 14 words. Details on the automated prompt expansion technique using a LLM, including the seed examples and generation procedure, are provided in Appendix A.

3.2. Experiment execution

We carried out a preliminary test to evaluate the influence of prompt semantic content on energy consumption. We selected as prompts the 100 distinct classes of the CIFAR-100 dataset, leveraging its broad class variety. Each model processed the prompts 10 times in randomized order to mitigate ordering biases. Then, to ensure consistency, all models were tested using their default settings, avoiding potential parameter-related influences.

The results of the preliminary test indicated no significant correlation between the semantic content of the prompts and energy

consumption, confirming that semantic variability is not a critical factor for this study. Further details on the preliminary test are provided in Appendix B.

Building upon the findings of the preliminary test, we conducted the main experiments of this study, which focused on measuring energy consumption for each model while varying the independent variables outlined in Section 3.1. To this end, we randomly selected 30 prompts from the CIFAR-100 dataset classes to form the short-length prompts. Then we expanded and truncated these prompts to obtain the long-length and medium-length prompt, following the approach detailed in Section 3.1.4. Similar to the preliminary experiment, to prevent biases associated to prompt ordering and boundary effects, the prompts were randomly shuffled before each experiment.

Our setup involved generating images using 17 different diffusion model under their default settings while varying two quantization parameters, three image resolution levels, and three prompt lengths. As a results, each prompt underwent $17 \times 2 \times 3 \times 3 = 540$ experimental runs, leading to a total of $540 \times 30 = 9180$ experiments across all prompts.

Energy consumption was measured via CodeCarbon (Courty et al., 2024), a Python library allowing to measure the energy consumed during the generation process of a image in kilowatt-hours (kWh) through the formula $E = E_{CPU} + E_{GPU} + E_{RAM}$, i.e., the sum of CPU, GPU, and RAM energy consumption. We manually set the CodeCarbon polling interval to 1s to ensure an adequate sampling resolution of the image generation energy consumption. The tracker was started immediately before each image-generation and was stopped immediately after. The collected values correspond to the energy estimated from CPU, GPU, and RAM usage during the end-to-end image generation stage of the Diffusers² pipeline. Specifically, monitoring started immediately before the generation call, after model loading and GPU warm-up, and stopped right after completion of the final denoising and VAE decoding steps. Model loading was intentionally excluded to ensure fair comparability across runs and configurations. To further mitigate potential initialization biases, each experiment was executed by repeating the first and the last prompt once before the actual measurement phase, acting as an additional warm-up step for the pipeline. These preliminary generations were then excluded from the reported measurements and from all subsequent analyses. Therefore, the reported values represent the total energy observed during the monitored execution interval, excluding

² We used Diffusers library by Hugging Face for generating images with the investigated diffusion models.

the first and the last measure. Regarding the conversion from kWh to carbon emission (CO_2e), we relied on the default setting provided by the CodeCarbon tool, which are based on the world-average carbon intensity of $475 \text{ gCO}_2/\text{kWh}$ (Courty et al., 2024).

All experiments were performed on a dedicated workstation with an AMD Ryzen 9 7950X processor, 128 GB of RAM, and an NVIDIA GeForce RTX 4090. More details about the hardware and software used can be found in Appendix C.

3.3. Image quality assessment

For image-quality evaluation, we considered both FID and the complementary Precision and Recall metrics. While FID provides an overall measure of the similarity between the generated and reference distributions, Precision and Recall allow us to disentangle the two main aspects of generative quality, namely fidelity and diversity, respectively. Computing these metrics requires a reference distribution of real images against which the generated images are compared.

Since no definitive reference set exists for text-to-image diffusion models, a common approach (Chen et al., 2023; Chadebec et al., 2024) is to use the validation set of the dataset from which class labels were used as prompts. In our setting, instead of using the CIFAR-100 validation images, which have low resolution (32×32), we employed the ImageNet-1k validation set. However, since the class labels in ImageNet-1k do not fully overlap with our selected prompt classes, we applied a sampling strategy to guarantee that the same subjects are uniformly represented. Specifically, we utilized our prompt classes as queries to identify corresponding classes in ImageNet-1k. First, we selected all ImageNet-1k classes which had at least one match with a prompt. If a unique match was found, we retrieved all 50 images for that class. Otherwise, where multiple matches existed (e.g., prompt “clock” matched multiple classes such as “wall clock”, “analog clock” and “digital clock”), we performed uniform random sampling across the matched classes to maintain balance and diversity in the final dataset. This procedure provides a representative and balanced real-image distribution for evaluating the quality of the generated samples.

For a detailed overview of the ImageNet classes used in the sampling process and their matching prompts, please refer to Table D.6 in Appendix D.

Via our filtering strategy we obtained a final dataset for quality assessment consisting of 1100 images from the ImageNet-1k validation set and 396 images for each diffusion model considered. FID computation was performed using IQA-PyTorch library (Chen and Mo, 2022), while Precision and Recall were computed using the implementation provided by Naeem et al. (2020).

4. Results

In this section, we present the results of our experiment by considering the various independent variables studied, i.e., models, quantization, image resolution, and prompt length. As concluding remark, we additionally explore the interplay between energy consumption and image quality.

4.1. Median model energy consumption

An overview of the overall cumulative energy consumption of the models (i.e., the median energy consumption of each model across all independent variables) is depicted in Fig. 1. As we can observe from Fig. 1, the considered models display a notable difference in terms of median energy consumption, with Lumina being the models consuming the highest energy value ($4.08 \times 10^{-3} \text{ kWh}$) and LCM_SSD_1B the lowest one ($8.6 \times 10^{-5} \text{ kWh}$). While the difference of median energy consumption is less drastic for other models (e.g., PixArt_Alpha and PixArt_Sigma differ only for a negligible 4×10^{-5} factor), we conclude

that choosing a model over another one can lead to drastic energy savings up to 46x. Overall, we can observe that U-Net-based models are generally more energy-efficient than Transformer-based architectures. This difference may be partly attributable to the lower computational and memory demands of convolutional operations compared to self-attention mechanisms; however, the evaluated models also differ in other aspects besides their architecture, which were not controlled for in this study and could influence energy consumption. Such considerations should be taken into account when interpreting these results. As detailed in Appendix E, Transformer-based models are generally slower at generating images compared to U-Net-based ones, which is also reflected in their higher usage of GPU, CPU, and RAM resources. We can further conjecture that architectural differences may interact with other factors influencing energy consumption, such as the number of diffusion steps required for image generation. In particular, models optimized for fewer inference steps (e.g., distilled or accelerated variants) can significantly reduce generation time and thus energy consumption, even when based on similar underlying architectures. As shown in Table E.7 in Appendix E, energy consumption is largely time-driven, with longer generation times typically corresponding to higher energy usage. This further supports the observation that approaches reducing the number of diffusion steps can substantially improve energy efficiency. As additional observation, we note that the variability in terms of energy consumption differs between models, with some displaying only a limited energy consumption fluctuation across settings (e.g., the Pixart-based models), while others a much higher one (e.g., Lumina, SD_1.5, and Flash_SD). Detailed results on average energy consumption per model, including component-wise estimates and average execution times, are reported in Appendix E.

Median Model Energy Consumption: Energy consumption varies drastically across models, showcasing up to a 46x increment. U-Net-based models tend to consume less than Transformer-based ones, possibly reflecting the greater computational complexity of transformer architectures among other factors, while consumption fluctuations vary from model to model.

4.2. Model quantization

Fig. 2 illustrates, on a logarithmic scale, the energy consumption (in kWh) of the analyzed models across three image resolutions, comparing their performance when quantization using the *int8* data type and *no quantization* is applied. The energy consumed values for each model are the medians across the 30 prompts. Additionally, Table 2 reports the percentage variation in energy consumption between the non-quantized and quantized versions of each model.

The results indicate that most models exhibit increased energy consumption when quantized to the *int8* data type, with the exception of four models—SDXL_Turbo, Hyper_SD, Flash_SD3, and Flux_1—which shows a modest reduction in energy usage, up to 11.16% (95% CI (Confidence Interval): 9.48%–13.37%). However, the percentage difference in energy consumption between the quantized and non-quantized versions is generally negligible, except for few models, e.g., SD_1.5, SDXL, and SSD_1B. This outcome is plausible as *int8* quantization may introduce additional convert-in and convert-out overheads on the GPU. This suggests that, in our setting, quantization does not improve the energy efficiency of the image generation process; however, its expected benefit in terms of memory usage was not directly measured in this study and therefore cannot be confirmed here. We note that these findings are specific to the evaluated setup, namely Quanto-based *int8* quantization on an NVIDIA RTX 4090 GPU. Different *int8* backends (e.g., TensorRT or bitsandbytes) or alternative hardware platforms may exhibit different energy–performance trade-offs.

Comprehensive energy consumption statistics for each model across quantization settings are provided in Appendix F.

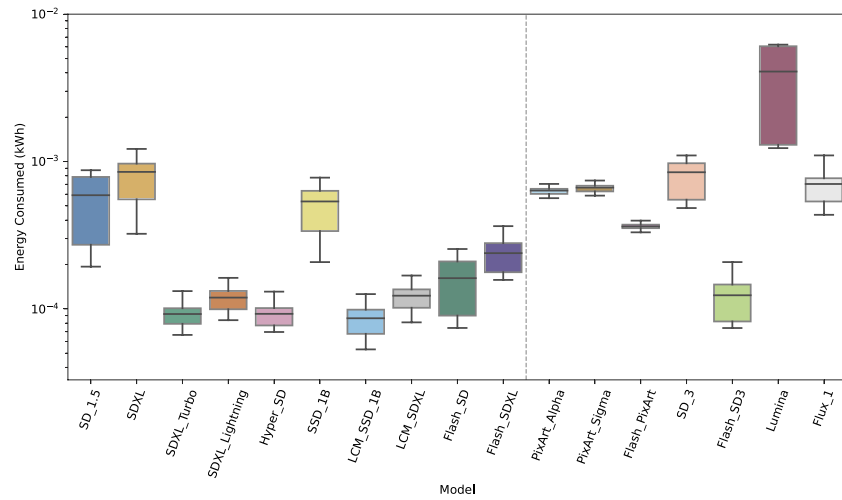


Fig. 1. Boxplot of energy consumption across the investigated models. The dotted gray lines separate U-Net (left) from Transformer-based models (right).

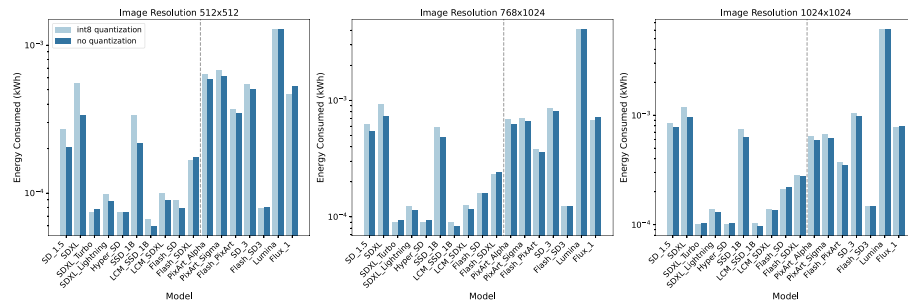


Fig. 2. Energy consumption (kWh) for image generation process of diffusion models at varying quantization levels. Bar charts (log scale) compare int8-quantized vs. non-quantized models at fixed resolutions. For all models, the default configuration corresponds to running without quantization. The dotted gray lines separate U-Net (left) from Transformer-based models (right). Each bar shows the median over 30 prompts.

Table 2

Percentage variation of energy consumption of diffusion models at varying quantization levels. Variation is computed as the difference between non-quantized and quantized models for each resolution. Negative values indicate lower energy consumption in non-quantized models. The reported 95% confidence intervals quantify the uncertainty associated with the estimated percentage variations, with narrower intervals indicating more precise estimates.

Type	Model	512 × 512		768 × 1024		1024 × 1024	
		Variation (%)	95% CI	Variation (%)	95% CI	Variation (%)	95% CI
U-Net-based	SD_1.5	-33.36	[-34.43, -32.57]	-13.79	[-14.59, -13.14]	-8.43	[-9.14, -7.79]
	SDXL	-64.68	[-65.85, -63.17]	-26.66	[-27.40, -26.14]	-22.36	[-22.92, -21.80]
	SDXL_Turbo	5.13	[4.11, 6.61]	2.16	[1.17, 3.56]	3.32	[2.27, 4.42]
	SDXL_Lightning	-11.83	[-13.31, -10.51]	-8.19	[-9.21, -7.13]	-6.00	[-7.22, -4.45]
	Hyper_SD	1.68	[0.00, 3.54]	3.43	[2.25, 4.62]	2.86	[1.93, 4.01]
	SSD_1B	-54.16	[-55.60, -52.74]	-22.11	[-22.81, -21.48]	-17.79	[-18.68, -16.79]
	LCM_SSD_1B	-11.46	[-13.00, -10.20]	-6.80	[-8.05, -5.62]	-5.59	[-7.00, -4.32]
	LCM_SDXL	-12.55	[-14.56, -10.73]	-7.89	[-9.13, -7.07]	-3.57	[-4.62, -2.61]
	Flash_SD	-13.36	[-14.89, -12.02]	0.92	[-2.86, 5.29]	4.98	[4.02, 5.89]
	Flash_SDXL	3.90	[2.48, 5.39]	2.57	[1.71, 3.20]	-1.74	[-2.89, -0.87]
Transformer-based	PixArt_Alpha	-8.34	[-9.37, -7.35]	-8.70	[-9.24, -8.04]	-8.02	[-8.86, -7.55]
	PixArt_Sigma	-8.95	[-10.14, -8.04]	-7.12	[-8.49, -6.25]	-9.72	[-10.51, -9.20]
	Flash_PixArt	-5.30	[-6.33, -4.34]	-6.37	[-7.11, -5.18]	-6.03	[-6.94, -5.26]
	SD_3	-8.33	[-9.63, -7.26]	-7.09	[-8.13, -6.31]	-7.68	[-8.73, -6.88]
	Flash_SD3	1.01	[-0.15, 1.98]	0.62	[-0.38, 1.75]	0.72	[-0.12, 1.51]
	Lumina	-0.13	[-0.73, 0.50]	-0.15	[-0.58, 0.16]	-0.21	[-0.63, 0.07]
	Flux_1	11.16	[9.48, 13.37]	5.63	[4.22, 7.00]	2.68	[0.61, 4.16]

Model Quantization: Within the investigated setting, model quantization leads in the vast majority of cases to an energy consumption increase (up to 64.7%) and only in rare cases to a negligible energy saving (up to 11.2%).

4.3. Image resolution

Fig. 3 shows the impact of image resolution on the energy consumption. Without any surprise, most of the investigated models exhibit the expected trend of increased energy consumption as image resolution increases. This can be explained by the greater amount of information

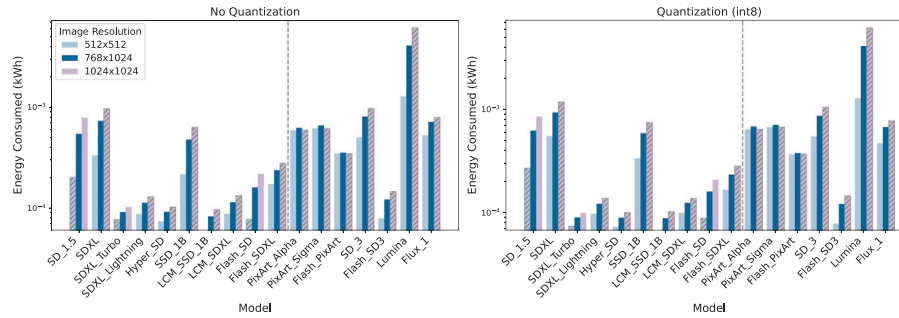


Fig. 3. Energy consumption (kWh) for image generation process of diffusion models at varying image resolution. Bar charts (log scale) compare 512×512 , 768×1024 , and 1024×1024 image resolutions at fixed quantization setting. Hatched bars (///) indicate the recommended image resolution for each model. The dotted gray lines separate U-net (left) from Transformer-based models (right). Each bar shows the median over 30 prompts.

that must be processed during image generation at higher resolutions, which amplifies the overall computational demand.

Table 3 highlights the varying energy consumption scaling of diffusion models as image resolution increases fourfold, from 512×512 to 1024×1024 . Energy consumption variation is calculated as the ratio of the energy used to generate a 1024×1024 resolution image to the energy used for 512×512 resolution image. Regarding U-Net-based models, some of them exhibit a relatively low scaling ratio (around 1.3), such as SDXL_Turbo and Hyper_SD, indicating a more efficient handling of higher resolutions. In contrast, models like SD_1.5 (3.85) and SSD_1B (2.90) show a much higher ratio, suggesting that their energy consumption is almost directly proportional to image resolution. For Transformer-based models, the PixArt family maintains a ratio close to 1.00, indicating that their energy consumption remains constant across resolutions. Other models, such as SD_3 and Flash_SD3, have slightly higher ratios (~ 1.9), implying a modest consumption increase. Lumina stands out with the highest ratio (4.75), exceeding even the expected quadratic scaling, which may suggest additional computational overhead or inefficiencies when processing high-resolution images. Detailed energy consumption statistics for each model, together with the percentage variation in energy consumption between the 512×512 and 1024×1024 image resolutions, are reported in [Appendix G](#). As peculiar result, the PixArt-based models showcased a higher energy consumption for non-square formats. Further experiments conducted by considering other 11 formats between the range of 256×256 to 1536×1536 confirmed this trend. We conjecture this trend is due to the training data used for the model, which could have been formatted only in square resolution. Additional details on the follow-up experiments are available in [Appendix G](#).

Image resolution: Image resolution affects differently the energy consumption of the models, with some showcasing a low scaling ratio (e.g., 1.3x when image resolution increases fourfold, from 512×512 to 1024×1024) while others show a considerable increase (e.g., 4.8x). Pixart family models remain close to constant across all square resolutions, albeit consuming considerably more for non-square ones.

4.4. Prompt length

As prompts with different lengths are derived from the same set of 30 original prompts, the resulting observations are not independent. Therefore, we assess whether prompt length affects energy consumption using the Friedman test, a non-parametric statistical test designed for repeated or matched measurements. Results (see [Appendix H](#) for more details) showcase that for all models the p-values are notably above the conventional significance level of 0.05, implying that prompt length is not a determining factor in the energy consumption of the examined models. This finding is consistent with the architecture of most

Table 3

Scaling ratios of energy consumption for image generation at 1024×1024 resolution compared to 512×512 resolution across different diffusion models.

Type	Model	No quantization	Quantization (int8)
U-Net-based	SD_1.5	3.85	3.13
	SDXL	2.88	2.14
	SDXL_Turbo	1.33	1.35
	SDXL_Lightning	1.48	1.41
	Hyper_SD	1.38	1.37
	SSD_1B	2.90	2.21
	LCM_SSD_1B	1.62	1.54
	LCM_SDXL	1.51	1.39
	Flash_SD	2.80	2.35
	Flash_SDXL	1.61	1.70
Transformer-based	PixArt_Alpha	1.00	1.00
	PixArt_Sigma	0.99	1.00
	Flash_PixArt	1.00	1.01
	SD_3	1.92	1.91
	Flash_SD3	1.85	1.86
	Lumina	4.75	4.75
	Flux_1	1.50	1.63

diffusion pipelines, where prompts are tokenized to a fixed maximum sequence length and then padded or truncated accordingly, resulting in a nearly constant computational cost independently of the original prompt length.

Prompt Length: Prompt length does not impact in a statistically significant manner the energy consumed by image generation.

4.5. Image quality assessment

Fig. 5 documents the performance of each diffusion model in terms of FID and energy consumption. Low values of FID correspond to better quality. Notably, Lumina emerges as the least efficient model, both from the quality and the energy perspective. On the other hand, Flash_SD3 offers the best compromise, achieving the lowest FID among all models while maintaining low energy consumption. As illustrated in [Fig. 4](#), where the same subject is represented at different resolutions, the final output of a diffusion model can vary significantly depending on the resolution. In particular, Lumina's high FID score is largely due to the poor quality of its 512×512 images, as the model is fine-tuned on resolutions of 1024×1024 and above, decreasing its ability to generate 512 -resolution images effectively. Similar effects are observed for other models, such as Flash_SD, SD_1.5 and SDXL_Turbo, which exhibit intermediate FID scores despite producing semantically altered images at 768×1024 and 1024×1024 resolutions.

FID captures both fidelity (realism) and diversity (variations in the original data) of generative models but struggles to separate their

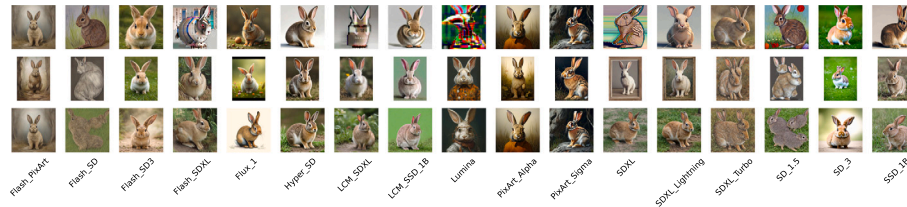


Fig. 4. Example images generated by each model at different resolutions. Each row corresponds to a different resolution: 512×512 (top), 768×1024 (middle), and 1024×1024 (bottom). The images were generated using the short prompt “rabbit” with int8 quantization.

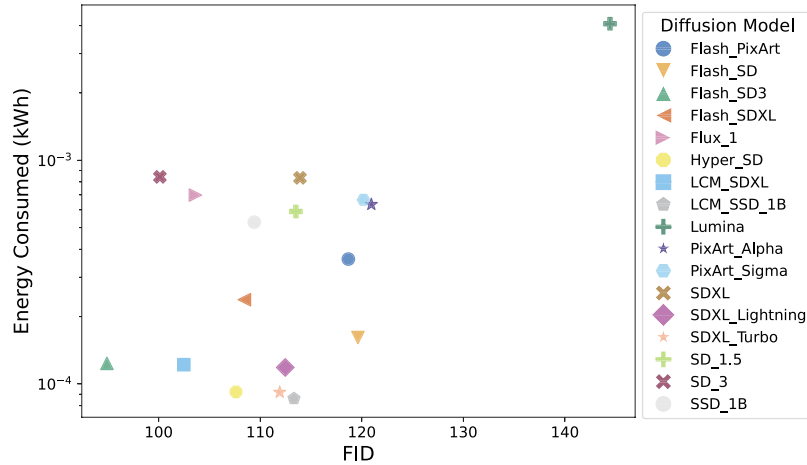


Fig. 5. Energy consumption (kWh) vs. FID scores for diffusion models. Y-axis is on a logarithmic scale. Each dot represents the median energy consumption per model.

contributions when FID scores are similar. Precision and Recall overcome this limitation by independently measuring fidelity and diversity, providing a more nuanced evaluation of image quality. In Fig. 6 each diffusion model is represented as a dot in the Precision–Recall space, with size indicating energy consumption. As we can observe, Lumina appears as the least precise model. Differently, while SD_1.5 and Hyper_SD achieve similar scores for FID in Fig. 5, their distinct positions in the Precision–Recall space highlight qualitative differences. SD_1.5 shows lower precision but higher recall with respect to Hyper_SD. This behavior can be seen as a drawback, as high recall often results from generating numerous diverse but unrealistic samples rather than effectively capturing subject heterogeneity (Naem et al., 2020). Based on this observation, Hyper_SD, SD_3, and Flash_SD3 stand out as the highest-quality models, combining strong performance with good energy efficiency.

Overall, these findings reveal a complex landscape in the evaluation of the image quality–energy trade off, contrary to the widespread assumption that higher image quality necessarily entails higher energy consumption. In particular, our results show that FID alone is insufficient to fully capture the relationship between quality and energy efficiency, as it conflates fidelity and diversity into a single score. By contrast, the joint analysis of Precision and Recall enables a clearer identification of highly precise models, as opposed to those achieving high diversity at the expense of realism. Furthermore, image resolution emerges as a critical factor in model comparison. Several models exhibit resolution-dependent energy–quality relationships (see Fig. D.11 and Fig. D.12 in Appendix D for resolution-specific plots), apparently linked to design choices, that make certain resolutions more natural and energy-efficient to generate than others, as in the case of Lumina. A comprehensive assessment of image quality–energy trade off therefore requires disentangled quality metrics and careful consideration of resolution effects in order to avoid potentially misleading conclusions about model efficiency. Additional discussion on CLIPScore variations across prompt lengths and its relationship with FID across models is provided in Appendix D.

Image Quality: Higher image quality does not come at the cost of higher energy consumption, with some of the most energy efficient models also producing among the highest quality images.

4.6. On the carbon emission associated to image generation

By considering the energy consumed by image generation models measured with our empirical experiment, we can gain a coarse figure of the carbon emission associated to generating an image via generative models. By considering the data provided by CodeCarbon,³ the date the experiment was executed (approximately in the nights of between the January 2 and January 3, 2025) and the location where the experiments were run approximated by the tool (namely Rome, Italy), we can gain an intuition of the CO₂ we emitted to generate the images in our experiment. At its peak, generating an image emitted up to 2.95g CO₂, measured when using the quantized Lumina model with a 1024×1024 resolution. While the median 0.160 g CO₂e (observed with the quantized SSD_1B model at 512×512 resolution) and minimum 0.025 g CO₂e (achieved by the non-quantized LCM_SSD_1B model at the same resolution) are overall lower figures, we note how, in a world where tens of millions images are generated each day,⁴ the global CO₂e associated to image generation is far from negligible, especially when less energy efficient models are used. With this research, we aim to set a first stepping stone to open the discourse on the environmental impact of generating images, in the hope it can foster further research, or even just create awareness, on the topic.

³ <https://codecarbon.io/>. Accessed 10 March 2026.

⁴ <https://journal.everypixel.com/ai-image-statistics>. Accessed 10 March 2026.

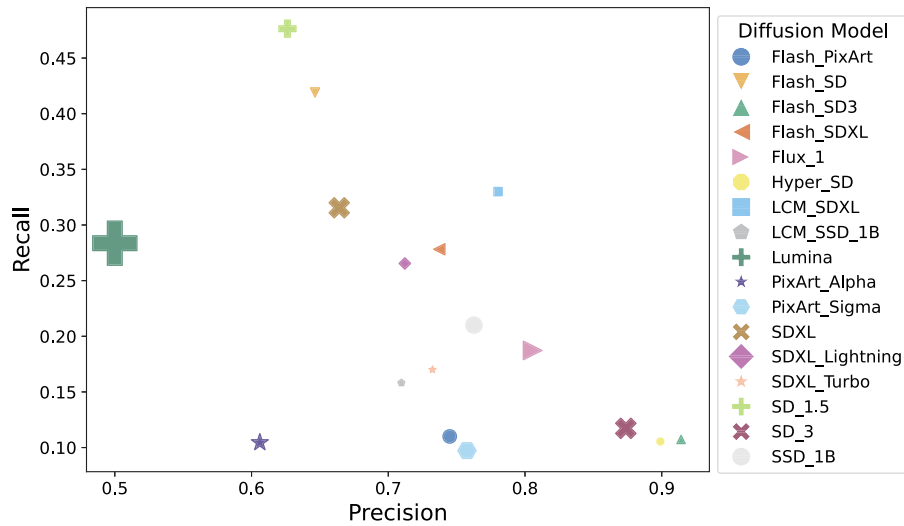


Fig. 6. Precision vs. Recall for diffusion models. Shape size is proportional to the median energy consumption per model.

5. Conclusions

In this study, we analyze the energy consumption of 17 diffusion models, considering multiple influencing factors at interplay. The measured energy varies significantly across models (up to a 46x increase), with U-Net-based models generally being more efficient than Transformer-based ones. Counterintuitively, model quantization often increases energy consumption (up to 64.68%), with only a few cases showing minor savings. Image resolution affects energy consumption inconsistently: A fourfold increase in resolution (from 512×512 to 1024×1024) produces energy increases ranging from 1.3x to 4.7x across models. Considering the quality-energy connection, Lumina ranks lowest, while Flash SD3 excels in both aspects.

All experiments in this study were conducted on a single hardware configuration (NVIDIA RTX 4090), which ensured consistent and controlled energy measurements across all evaluated models and settings. However, the absolute energy values reported may vary across different hardware platforms. For instance, modern data-center GPUs such as the NVIDIA H100 provide higher memory bandwidth, more advanced tensor cores, and support for lower-precision formats (e.g., FP8), which may improve both throughput and energy efficiency in large-scale inference scenarios. Although peak power consumption and scaling behavior may differ across architectures, we expect the qualitative trends observed in this work to remain largely consistent. Extending this analysis to heterogeneous hardware environments therefore represents a promising direction for future research and would further strengthen the generalizability of the findings. We also acknowledge the lack of reproducibility checks on the same hardware configuration, as the GPU used in our experiments is no longer commercially available.

In conclusion, with this research, we present a fine-grained empirical perspective on the energy required to generate images. With our study, we hope to have set a stepping stone to progress image generation not only by producing more eye pleasing images, but also by improving the environmental impact hidden behind each image we generate.

CRedit authorship contribution statement

Giulia Bertazzini: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Chiara Albisani:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Daniele Baracchi:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Dasara Shullani:** Writing – review & editing,

Writing – original draft, Supervision, Methodology, Conceptualization. **Roberto Verdecchia:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Prompt expansion through LLM

We used a large language model (LLM), i.e. Phi-3-Mini-4 K-Instruct (Abdin et al., 2024), to lengthen prompts rather than a neutral hand template in order to ensure objectivity, scalability, and reproducibility. This approach allowed us to automatically generate prompts that were both meaningful and diverse, avoiding the potential limitations of human-generated prompts, which may be more constrained in phrasing. The automated expansion produces prompts that are realistic and detailed enough to generate images with recognizable semantic content, reflecting realistic use cases.

The LLM is provided with several pairs of short prompt–prompt expansions, listed in Table A.4, which serve as seed examples, and is subsequently tasked with generating an expansion for a novel prompt as follows:

“You are a helpful AI assistant designed to help the user produce prompts that will be used to generate images using Stable Diffusion. Help the user by modifying a given prompt, adding many details so that the generated images will look better. Generate only realistic prompts. Do not engage in artistic and unrealistic prompts. Please improve this prompt: {prompt}”

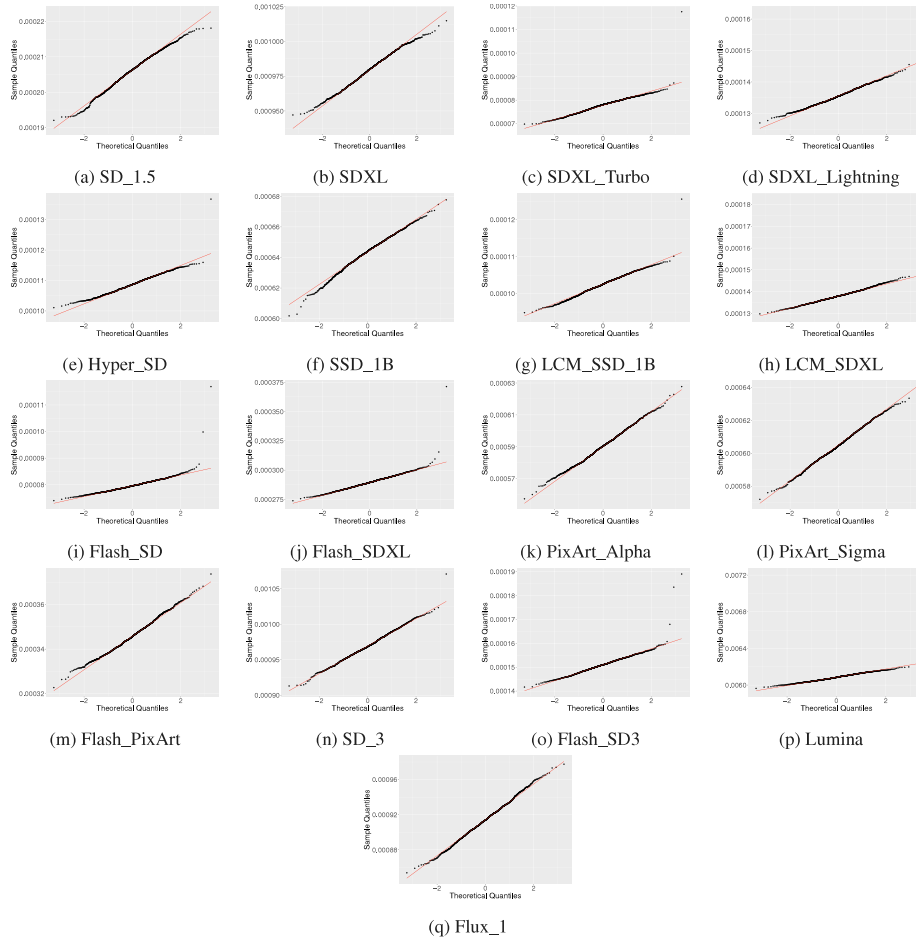
For each prompt, we used the default decoding settings (temperature = 0.8, top_p = 0.9, max_tokens = -1, i.e., no limit, and a random seed), chosen to balance diversity and semantic coherence in the generated expansions.

We also performed a sanity check on all 30 used prompts, performed independently by two annotators, to ensure that expansion preserves the original class meaning. With this test, we found agreement on 29 out of 30 prompts (96.7% agreement), corresponding to an inter-rater agreement measured with Cohen’s $\kappa \approx 0.933$, indicating nearly perfect agreement. The only disagreement was on the prompt “ray”, which was expanded in “ray of light streaming through the window in a cozy living room setting, morning scene, warm glow on wooden furniture, natural ambient lighting, minimalist decor, vintage lifestyle photography, soft focus, bokeh effect, Canon EOS 5D Mark IV”, due to the word’s double meaning.

Table A.4

Short prompt–prompt expansions pairs given as seed examples to Phi-3.

Short prompt	Prompt expansion
a portrait of a blonde woman	portrait of a pretty blonde woman, a flower crown, earthy makeup, flowing maxi dress with colorful patterns and fringe, a sunset or nature scene, green and gold color scheme
a portrait of an old man	photorealistic, visionary portrait of a dignified older man with weather-worn features, digitally enhanced, high contrast, chiaroscuro lighting technique, intimate, close-up, detailed, steady gaze, rendered in sepia tones, evoking rembrandt, timeless, expressive, highly detailed, sharp focus, high resolution
a realistic picture of a living room	a living room, bright modern Scandinavian style house, large windows, magazine photoshoot, 8k, studio lighting
a closeup of a goth woman	closeup portrait photo of beautiful goth woman, makeup, 8k uhd, high quality, dramatic, cinematic
a picture of a rabbit in a forest	close up photo of a rabbit, forest in spring, haze, halation, bloom, dramatic atmosphere, centered, rule of thirds, 200 mm 1.4f macro shot
a professional picture of a luxury bag	brehtaking shot of a bag, luxury product style, elegant, sophisticated, high-end, luxurious, professional, highly detailed
johnny depp, noir style	johnny depp photo portrait, film noir style, monochrome, high contrast, dramatic shadows, 1940s style, mysterious, cinematic
tokyo, long exposure	long exposure photo of tokyo street, blurred motion, streaks of light, surreal, dreamy, ghosting effect, highly detailed
a photoshoot of a model, cyberpunk style	a glamorous digital magazine photoshoot, a fashionable model wearing avant-garde clothing, set in a futuristic cyberpunk roof-top environment, with a neon-lit city background, intricate high fashion details, backlit by vibrant city glow, Vogue fashion photography
floral tea	freshly made hot floral tea in glass kettle on the table, angled shot, midday warm, Nikon D850 105 mm, close-up
a picture of a smiling girl, red hair, upper body shot	masterpiece, best quality, girl, collarbone, wavy hair, looking at viewer, blurry foreground, upper body, necklace, contemporary, plain pants, intricate, print, pattern, ponytail, freckles, red hair, dappled sunlight, smile, happy

**Fig. B.7.** Q-Q plots comparing the energy consumption of the analyzed models across various prompt contents to the normal distribution. Each plot visualizes the distribution of energy consumption values for a specific model, assessing how closely they follow a normal distribution.

Appendix B. Preliminary experiment

The results of the correlation between the semantic content of prompts and the energy consumed during the image generation process are presented in Table B.5. Before performing the correlation analysis, we verified that the energy consumption for each model followed a normal distribution, by examining Quantile–Quantile (QQ) plots, which are illustrated in Fig. B.7. If the data closely follows the normal distribution, the points on the Q-Q plot will move on a diagonal line (depicted in red), while deviations from the reference line indicate departures from the expected distribution. Fig. B.7 shows that for all the analyzed models, data follows a normal distribution.

See Figs. B.8–B.10.

From Table B.5, we can observe that, with the exception of SD_3, for none of models the ANOVA test reached statistical significance.

These findings indicate that there is no statistically significant relationship between the prompt content and the energy consumed during the generation process, supporting the validity of the followed experimental setup.

Additionally, Figs. B.8–B.10 show the box plots which illustrate the distribution of energy consumption across different models for the 100 different prompts. As we can see, the median energy remains quite stable across all semantics, with negligible variations among prompts.

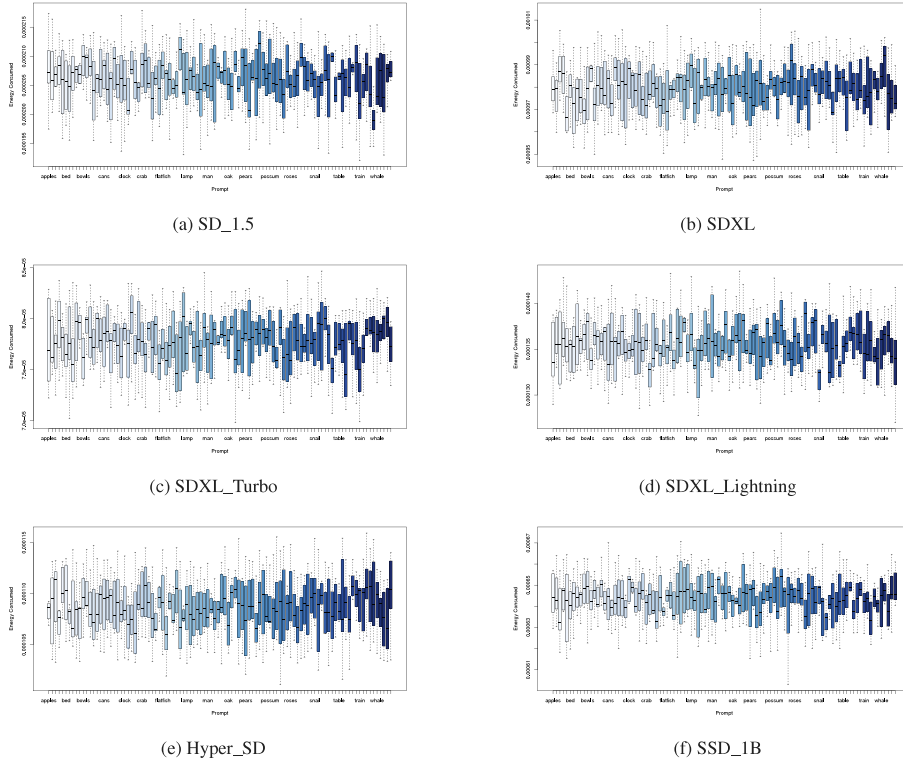


Fig. B.8. Box plots of the energy consumption across various prompt contents for the analyzed models. (Group 1).

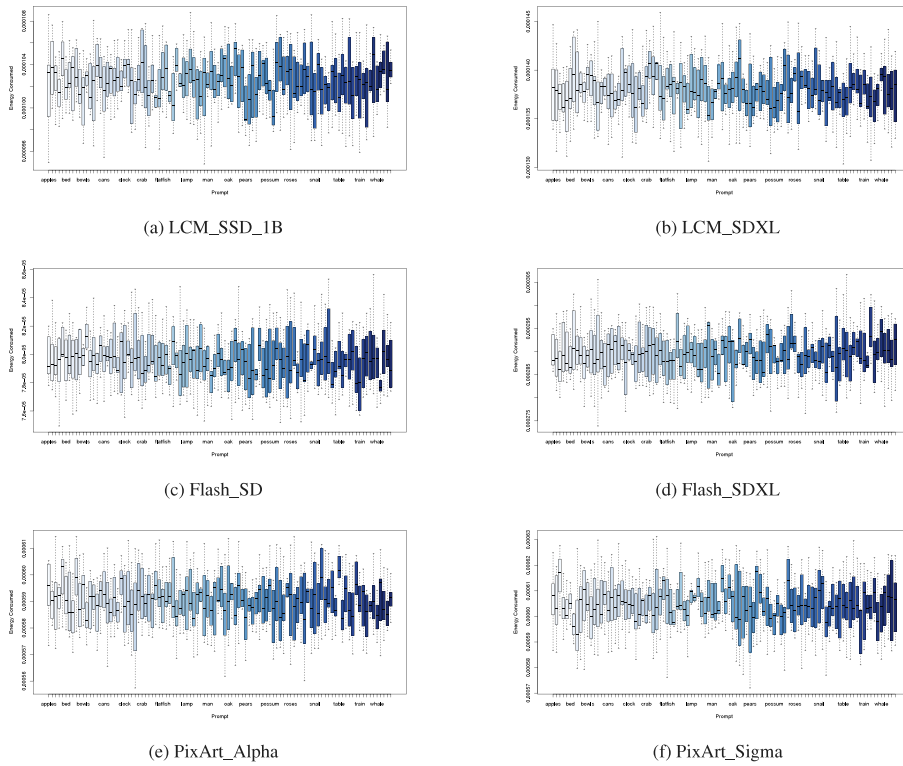


Fig. B.9. Box plots of the energy consumption across various prompt contents for the analyzed models. (Group 2).

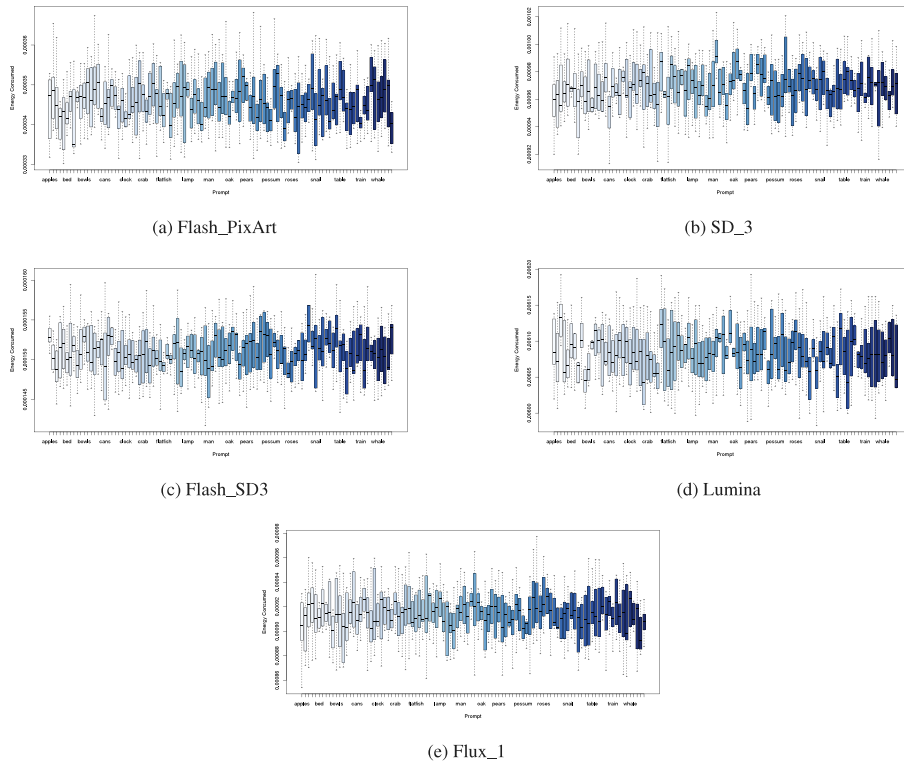


Fig. B.10. Box plots of the energy consumption across various prompt contents for the analyzed models. (Group 3).

Table B.5

One-way ANOVA p-values for the effect of prompt topic on energy consumption across the evaluated models.

Model	P-value
Flash_PixArt	0.373
Flash_SD	0.958
Flash_SD3	0.184
Flash_SDXL	0.715
Flux_1	0.940
Hyper_SD	0.725
LCM_SDXL	0.798
LCM_SSD_1B	0.457
Lumina	0.129
PixArt_Alpha	0.656
PixArt_Sigma	0.855
SDXL	0.820
SDXL_Lightning	0.074
SDXL_Turbo	0.084
SD_1.5	0.860
SD_3	0.030
SSD_1B	0.878

This observation supports the finding from ANOVA test that prompt semantic does not impact on the energy consumption.

Appendix C. Hardware and software details

All experiments were conducted on a system equipped with an AMD Ryzen 9 7950X processor, 128 GB of RAM, NVIDIA GeForce RTX 4090 GPU with a total graphics power (TGP) of 450 W. The RTX 4090 was selected as a widely available high-end consumer GPU representative of the hardware commonly used for local deployment of

diffusion models. The GPU operated using default persistence mode and clock settings. The system ran Ubuntu 22.04.5 LTS with NVIDIA driver 565.x and CUDA 12.3. The software stack included CodeCarbon 2.8.1, PyTorch 2.5.1, Diffusers 0.31.0, and Transformers 4.47.1. While this setup provides a consistent and representative hardware configuration for our experiments, we acknowledge that the measured energy consumption may vary across different GPU architectures and hardware configurations.

Appendix D. Quality assessment

As previously stated in Section 3.3, the validation set of ImageNet-1k was properly filtered to ensure that only the classes matching with the prompt labels were retained, with random uniform sampling in case of multiple matches. Unfortunately, for prompts *apples*, *girl*, *poppies*, *aquarium fish*, *oak*, *shrew*, *ray* and *tulips* were not possible to find a matching ImageNet-1k class. Consequently, we excluded all images associated with the aforementioned prompts from the calculation of the quality metrics.

Figs. D.11 and D.12 illustrate how the quality-energy relationship varies when considering images at specific resolutions separately. Flash_SD3 maintains consistent FID and Precision-Recall performance across different resolutions, whereas many other models exhibit significant variations. Among them, Flash_SD shoes low FID combined with high precision at 512×512 but shifts to higher FID, lower precision and higher recall at 768×1024 and 1024×1024 . These trends are visually confirmed by the generated images in Figs. D.13, D.14 and D.15.

Fig. D.15 presents the same subject (prompt number 0) across different diffusion models at 1024×1024 resolution with int8 quantization, varying only the prompt length. As discussed in Appendix H, prompt length generally does not impact the energy consumption of the generation process. From a quality perspective, short-prompt images cannot be definitively classified as lower quality. However, it is

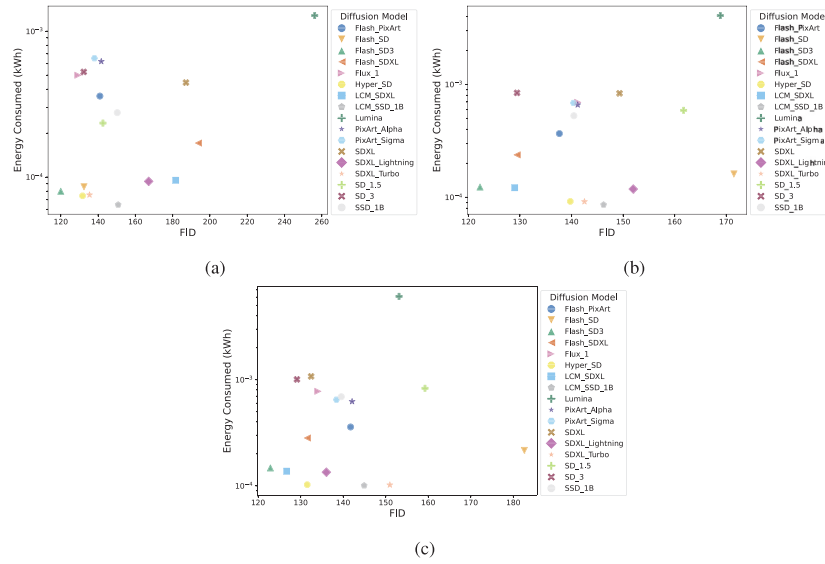


Fig. D.11. Energy consumption (kWh) vs FID values for all the models at resolution (a) 512×512 (b) 768×1024 and (c) 1024×1024 .

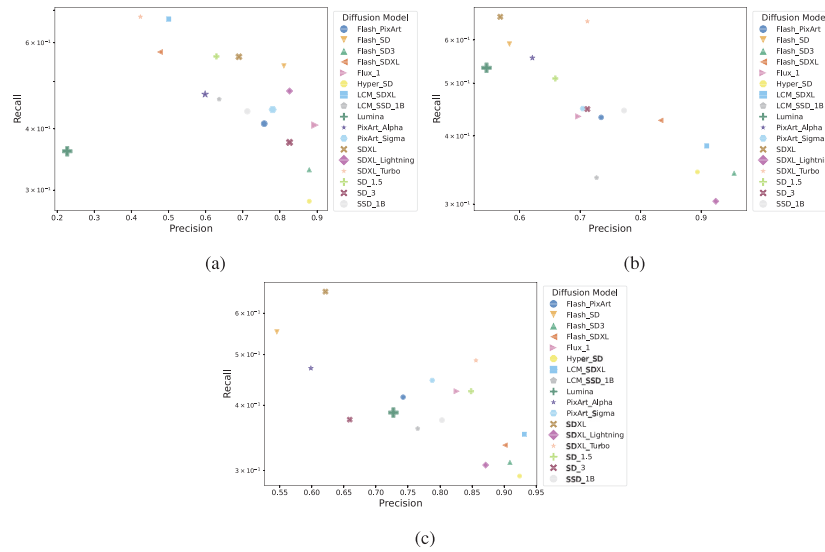


Fig. D.12. Precision and Recall values for each model at resolution (a) 512×512 (b) 768×1024 and (c) 1024×1024 . Each symbol of the plot has different size based on the energy consumed.



Fig. D.13. Example images generated by each model at resolution 512×512 , quantization int8 and prompt “bear”. Each row corresponds to a different prompt length: long (top), medium (middle), and short (bottom).



Fig. D.14. Example images generated by each model at resolution 768×1024 , quantization int8 and prompt “tree”. Each row corresponds to a different prompt length: long (top), medium (middle), and short (bottom).

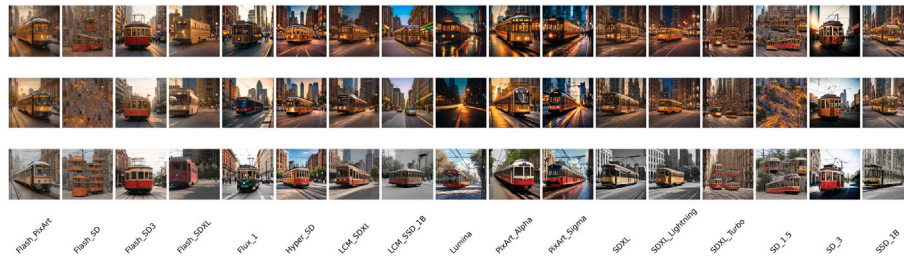


Fig. D.15. Example images generated by each model at resolution 1024×1024 , quantization int8 and prompt “streetcar”. Each row corresponds to a different prompt length: long (top), medium (middle), and short (bottom).

interesting to note that the most significant differences in image content arise between short and long prompts. This aligns with the goal of prompt engineering for text-to-image generation, which is to produce more detailed and complex images. Other examples are shown in Figs. D.13 and D.14.

Fig. D.16 presents the average CLIPScore values for all examined models across varying prompt lengths. The results indicate that shorter prompts consistently achieve the highest scores for every model, whereas longer prompts yield the lowest. This pattern suggests that when prompts contain only limited information, the generated images are less likely to deviate from the requested content, resulting in looser alignment with the given text. In contrast, longer prompts introduce more details, which the generation process may struggle to accurately capture, leading to greater misalignment, and consequently lower CLIPScore values. We observe that, differently from FID, CLIPScore values are relatively similar across models. This suggests that the examined models achieve comparable levels of text-image alignment, whereas visual realism differs more markedly. This supports our choice of focusing on FID when studying the relationship between image quality and energy consumption in our setting.

Appendix E. Model energy consumption

Table E.7 reports the average image generation time per image, together with the average energy consumption and average power, the latter computed as energy divided by execution time. These metrics are reported both as cumulative values and disaggregated by hardware component, namely CPU, GPU, and RAM, including component-wise energy consumption and the corresponding average power estimates. This analysis indicates that the observed differences are primarily driven by execution time. Indeed, the computational time required to generate an image is highly model-dependent: while models such as LCM_SSD_1B or Hyper_SD require approximately 1 s per image, generation can take up to 34 s when using Lumina. Overall, U-Net-based models tend to generate images faster than Transformer-based architectures.

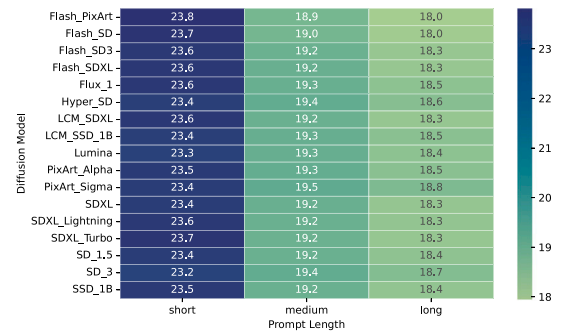


Fig. D.16. Average CLIPScore values for each diffusion model with respect to different prompt lengths.

Appendix F. Model quantization

Tables F.8–F.10 present detailed descriptive statistics of energy consumption (in kWh) for each model across three image resolutions, comparing their performance under int8 quantization versus no quantization. For each configuration, the minimum, maximum, mean, median, standard deviation, and interquartile range (IQR) are reported.

See Tables F.8–F.10.

Appendix G. Image resolution

Tables G.11 and G.12 present detailed descriptive statistics of energy consumption (in kWh) for each model across two quantization settings (i.e., int8 quantization and no quantization), comparing their performance under different image resolution levels. For each configuration, the minimum, maximum, mean, median, standard deviation, and interquartile range (IQR) are reported.

Table D.6

List of ImageNet class_id, class_label that intersect with our selected prompts. Corresponding number of images for each class are reported on the last column.

ImageNet_class_id	ImageNet_class_label	Prompt	#Images
n02134418	sloth bear, Melursus ursinus, Ursus ursinus	bear	8
n04399382	teddy, teddy bear	bear	8
n02134084	ice bear, polar bear, Ursus Maritimus, Thalarctos maritimus	bear	8
n02133161	American black bear, black bear, Ursus americanus, Euarctos americanus	bear	8
n02132136	brown bear, bruin, Ursus arctos	bear	9
n01882714	koala, koala bear, kangaroo bear, native bear, Phascolarctos cinereus	bear	9
n04366367	suspension bridge	bridge	25
n04311004	steel arch bridge	bridge	25
n02279972	monarch, monarch butterfly, milkweed butterfly, Danaus plexippus	butterfly	10
n02277742	ringlet, ringlet butterfly	butterfly	10
n02281787	lycaenid, lycaenid butterfly	butterfly	10
n02280649	cabbage butterfly	butterfly	10
n02281406	sulfur butterfly, sulfur butterfly	butterfly	10
n02481823	chimpanzee, chimp, Pan troglodytes	chimpanzee	50
n04548280	wall clock	clock	16
n02708093	analog clock	clock	17
n03196217	digital clock	clock	17
n03085013	computer keyboard, keypad	computer keyboard	50
n04344873	studio couch, day bed	couch	50
n02504013	Indian elephant, Elephas maximus	elephant	25
n02504458	African elephant, Loxodonta africana	elephant	25
n02119022	red fox, Vulpes vulpes	fox	13
n02120079	Arctic fox, white fox, Alopex lagopus	fox	12
n02120505	gray fox, gray fox, Urocyon cinereoargenteus	fox	12
n02119789	kit fox, Vulpes macrotis	fox	13
n02797295	barrow, garden cart, lawn cart, wheelbarrow	lawn-mower	25
n03649909	lawn mower, mower	lawn-mower	25
n02128385	leopard, Panthera pardus	leopard	25
n02128757	snow leopard, ounce, Panthera uncia	leopard	25
n02125311	cougar, puma, catamount, mountain lion, painter, panther, Felis concolor	lion	25
n02129165	lion, king of beasts, Panthera leo	lion	25
n07747607	black	blacks	50
n03961711	plate rack	plates	25
n07579787	plate	plates	25
n02325366	wood rabbit, cottontail, cottontail rabbit	rabbit	25
n02328150	Angora, Angora rabbit	rabbit	25
n01773157	black and gold garden spider, Argiope aurantia	spider	13
n01773797	garden spider, Aranea diademata	spider	12
n01773549	barn spider, Araneus cavaticus	spider	13
n01775062	wolf spider, hunting spider	spider	12
n04335435	streetcar, tram, tramcar, trolley, trolley car	streetcar	50
n04389033	tank, army tank, armored combat vehicle, armored combat vehicle	tank	50
n02992529	cellular telephone, cellular phone, cellphone, cell, mobile phone	telephone	25
n03187595	dial telephone, dial phone	telephone	25
n02123159	tiger cat	tiger	25
n02129604	tiger, Panthera tigris	tiger	25
n01669191	box turtle, box tortoise	turtle	10
n01667778	terrapin	turtle	10
n01667114	mud turtle	turtle	10
n01665541	leatherback turtle, leatherback, leathery turtle, Dermochelys coriacea	turtle	10
n01664065	loggerhead, loggerhead turtle, Caretta caretta	turtle	10
n01924916	flatworm, platyhelminth	worm	17
n01930112	nematode, nematode worm, roundworm	worm	16
n01728572	thunder snake, worm snake, Carphophis amoenus	worm	17

Table E.7

Average execution time, energy consumption, and power per generated image for all evaluated models. Reported values include both cumulative measurements and component-wise estimates provided by CodeCarbon, disaggregated into CPU, GPU, and RAM contributions. Energy is measured over the end-to-end image generation process (excluding model loading), while average power is computed as the ratio between energy and execution time. E. denotes energy and P. denotes power.

Type	Model	Time/img (s)	Energy (kWh)	Power (kW)	CPU E. (kWh)	GPU E. (kWh)	RAM E. (kWh)	CPU P. (W)	GPU P. (W)	RAM P. (W)
U-Net-based	SD_1.5	4.98	5.49e-04	3.97e-01	9.18e-05	3.92e-04	6.47e-05	6.63e+01	2.68e+02	4.69e+01
	SDXL	7.66	7.92e-04	3.72e-01	1.41e-04	5.52e-04	9.98e-05	6.64e+01	2.57e+02	4.69e+01
	SDXL_Turbo	1.63	9.82e-05	2.17e-01	3.03e-05	4.68e-05	2.12e-05	6.76e+01	1.07e+02	4.69e+01
	SDXL_Lightning	1.78	1.19e-04	2.41e-01	3.32e-05	6.27e-05	2.31e-05	6.72e+01	1.27e+02	4.69e+01
	Hyper_SD	1.51	9.26e-05	2.21e-01	2.83e-05	4.48e-05	1.96e-05	6.73e+01	1.07e+02	4.69e+01
	SSD_1B	4.93	5.05e-04	3.68e-01	9.08e-05	3.50e-04	6.43e-05	6.62e+01	2.51e+02	4.69e+01
	LCM_SSD_1B	1.36	8.99e-05	2.38e-01	2.52e-05	4.71e-05	1.77e-05	6.72e+01	1.28e+02	4.69e+01
	LCM_SDXL	1.90	1.25e-04	2.37e-01	3.53e-05	6.50e-05	2.47e-05	6.72e+01	1.26e+02	4.69e+01
	Flash_SD	2.00	1.59e-04	2.86e-01	3.65e-05	9.64e-05	2.61e-05	6.59e+01	1.70e+02	4.69e+01
	Flash_SDXL	3.54	2.45e-04	2.49e-01	6.50e-05	1.34e-04	4.62e-05	6.63e+01	1.38e+02	4.69e+01
Transformer-based	PixArt_Alpha	8.20	6.36e-04	2.79e-01	1.50e-04	3.79e-04	1.07e-04	6.58e+01	1.67e+02	4.69e+01
	PixArt_Sigma	8.48	6.64e-04	2.82e-01	1.56e-04	3.98e-04	1.10e-04	6.60e+01	1.70e+02	4.69e+01
	Flash_PixArt	6.37	3.64e-04	2.06e-01	1.16e-04	1.66e-04	8.29e-05	6.53e+01	9.37e+01	4.69e+01
	SD_3	10.05	8.05e-04	2.89e-01	1.84e-04	4.90e-04	1.31e-04	6.60e+01	1.73e+02	4.69e+01
	Flash_SD3	1.72	1.24e-04	2.59e-01	3.18e-05	6.96e-05	2.24e-05	6.65e+01	1.45e+02	4.69e+01
	Lumina	33.97	3.82e-03	4.05e-01	6.12e-04	2.77e-03	4.42e-04	6.56e+01	2.81e+02	4.69e+01
	Flux_1	10.77	7.03e-04	2.35e-01	1.97e-04	3.65e-04	1.40e-04	6.62e+01	1.25e+02	4.69e+01

Table F.8

Energy consumption (in kWh) statistics for each model at **fixed 512 × 512 resolution**, comparing INT8 quantization and no quantization runs. Reported values include minimum, maximum, mean, median, standard deviation, and interquartile range (IQR), providing a detailed view of variability across prompts.

Type	Model	Quantization	Min	Max	Mean	Median	Std	IQR
U-Net-based	SD_1.5	int8	2.54e-04	2.89e-04	2.72e-04	2.72e-04	6.89e-06	7.43e-06
		no quantization	1.93e-04	2.15e-04	2.04e-04	2.04e-04	4.79e-06	6.09e-06
	SDXL	int8	5.32e-04	5.78e-04	5.54e-04	5.54e-04	9.55e-06	1.52e-05
		no quantization	3.23e-04	3.84e-04	3.38e-04	3.36e-04	1.02e-05	1.08e-05
	SDXL_Turbo	int8	6.65e-05	9.66e-05	7.48e-05	7.45e-05	5.10e-06	3.77e-06
		no quantization	7.08e-05	1.16e-04	7.92e-05	7.84e-05	7.58e-06	4.24e-06
	SDXL_Lightning	int8	9.28e-05	1.16e-04	1.00e-04	9.91e-05	4.80e-06	5.49e-06
		no quantization	8.37e-05	1.27e-04	8.96e-05	8.87e-05	6.77e-06	2.96e-06
	Hyper_SD	int8	6.96e-05	8.68e-05	7.44e-05	7.42e-05	3.34e-06	2.49e-06
		no quantization	7.01e-05	1.04e-04	7.63e-05	7.53e-05	6.08e-06	5.17e-06
	SSD_1B	int8	3.22e-04	3.71e-04	3.38e-04	3.37e-04	7.84e-06	7.54e-06
		no quantization	2.08e-04	2.49e-04	2.19e-04	2.18e-04	6.40e-06	8.15e-06
	LCM_SSD_1B	int8	6.32e-05	8.49e-05	6.83e-05	6.74e-05	3.92e-06	2.60e-06
		no quantization	5.30e-05	8.81e-05	6.13e-05	6.05e-05	5.66e-06	3.31e-06
	LCM_SDXL	int8	9.44e-05	1.16e-04	1.01e-04	1.01e-04	4.14e-06	5.16e-06
		no quantization	8.09e-05	1.25e-04	9.06e-05	8.94e-05	7.23e-06	4.24e-06
	Flash_SD	int8	8.49e-05	9.61e-05	8.97e-05	8.94e-05	2.69e-06	3.68e-06
		no quantization	7.41e-05	1.18e-04	8.09e-05	7.88e-05	8.02e-06	3.19e-06
	Flash_SDXL	int8	1.57e-04	1.84e-04	1.68e-04	1.68e-04	6.86e-06	1.10e-05
		no quantization	1.59e-04	2.68e-04	1.82e-04	1.74e-04	2.26e-05	1.07e-05
Transformer-based	PixArt_Alpha	int8	6.23e-04	6.70e-04	6.43e-04	6.42e-04	1.12e-05	1.56e-05
		no quantization	5.68e-04	6.19e-04	5.93e-04	5.93e-04	1.17e-05	1.73e-05
	PixArt_Sigma	int8	6.52e-04	7.05e-04	6.75e-04	6.75e-04	1.13e-05	1.58e-05
		no quantization	5.90e-04	6.54e-04	6.20e-04	6.20e-04	1.37e-05	1.87e-05
	Flash_PixArt	int8	3.56e-04	3.84e-04	3.69e-04	3.68e-04	6.74e-06	9.50e-06
		no quantization	3.31e-04	3.68e-04	3.50e-04	3.50e-04	7.57e-06	1.04e-05
	SD_3	int8	5.19e-04	5.89e-04	5.49e-04	5.48e-04	1.44e-05	2.14e-05
		no quantization	4.83e-04	6.28e-04	5.10e-04	5.06e-04	2.09e-05	1.55e-05
	Flash_SD3	int8	7.43e-05	1.23e-04	8.28e-05	7.94e-05	1.04e-05	3.92e-06
		no quantization	7.41e-05	1.26e-04	8.29e-05	8.03e-05	1.08e-05	3.61e-06
	Lumina	int8	1.23e-03	1.37e-03	1.29e-03	1.28e-03	2.99e-05	3.31e-05
		no quantization	1.24e-03	1.36e-03	1.28e-03	1.28e-03	2.68e-05	3.29e-05
	Flux_1	int8	4.35e-04	5.37e-04	4.79e-04	4.74e-04	2.54e-05	4.10e-05
		no quantization	4.55e-04	8.51e-04	5.65e-04	5.31e-04	9.24e-05	9.68e-05

Table F.9

Energy consumption statistics (in kWh) for each model at **fixed 768×1024 resolution**, comparing INT8 quantization and no quantization runs. Reported values include minimum, maximum, mean, median, standard deviation, and interquartile range (IQR), providing a detailed view of variability across prompts.

Type	Model	Quantization	Min	Max	Mean	Median	Std	IQR
U-Net-based	SD_1.5	int8	6.05e-04	6.42e-04	6.25e-04	6.26e-04	8.28e-06	1.24e-05
		no quantization	5.25e-04	5.76e-04	5.50e-04	5.51e-04	1.04e-05	1.40e-05
	SDXL	int8	9.08e-04	9.65e-04	9.34e-04	9.32e-04	1.13e-05	1.52e-05
		no quantization	7.04e-04	7.93e-04	7.36e-04	7.36e-04	1.46e-05	1.84e-05
	SDXL_Turbo	int8	8.64e-05	1.17e-04	9.19e-05	9.11e-05	5.28e-06	3.76e-06
		no quantization	8.69e-05	1.32e-04	9.48e-05	9.30e-05	7.76e-06	4.05e-06
	SDXL_Lightning	int8	1.15e-04	1.45e-04	1.24e-04	1.24e-04	5.33e-06	4.21e-06
		no quantization	1.09e-04	1.51e-04	1.16e-04	1.15e-04	6.77e-06	4.45e-06
	Hyper_SD	int8	8.58e-05	1.06e-04	9.15e-05	9.06e-05	3.91e-06	3.53e-06
		no quantization	8.78e-05	1.25e-04	9.50e-05	9.38e-05	6.25e-06	3.14e-06
	SSD_1B	int8	5.61e-04	6.17e-04	5.88e-04	5.90e-04	1.01e-05	1.41e-05
		no quantization	4.58e-04	5.10e-04	4.83e-04	4.82e-04	8.77e-06	9.49e-06
	LCM_SSD_1B	int8	8.26e-05	1.12e-04	9.02e-05	8.95e-05	4.42e-06	4.06e-06
		no quantization	7.96e-05	1.09e-04	8.48e-05	8.41e-05	4.79e-06	2.12e-06
	LCM_SDXL	int8	1.18e-04	1.48e-04	1.27e-04	1.26e-04	5.82e-06	5.12e-06
		no quantization	1.07e-04	1.56e-04	1.18e-04	1.16e-04	7.79e-06	5.52e-06
Transformer-based	Flash_SD	int8	1.46e-04	1.72e-04	1.60e-04	1.61e-04	7.18e-06	1.15e-05
		no quantization	1.48e-04	2.10e-04	1.65e-04	1.63e-04	1.30e-05	1.85e-05
	Flash_SDXL	int8	2.22e-04	2.52e-04	2.36e-04	2.35e-04	6.19e-06	7.06e-06
		no quantization	2.27e-04	3.13e-04	2.46e-04	2.41e-04	1.85e-05	7.49e-06
	PixArt_Alpha	int8	6.63e-04	7.04e-04	6.85e-04	6.86e-04	8.58e-06	1.20e-05
		no quantization	6.06e-04	6.59e-04	6.32e-04	6.31e-04	1.12e-05	1.47e-05
	PixArt_Sigma	int8	6.85e-04	7.43e-04	7.12e-04	7.10e-04	1.37e-05	2.13e-05
		no quantization	6.39e-04	6.89e-04	6.62e-04	6.63e-04	1.02e-05	1.67e-05
	Flash_PixArt	int8	3.65e-04	3.97e-04	3.80e-04	3.80e-04	7.55e-06	1.02e-05
		no quantization	3.41e-04	3.75e-04	3.58e-04	3.57e-04	6.64e-06	8.60e-06
	SD_3	int8	8.34e-04	9.20e-04	8.71e-04	8.68e-04	1.74e-05	2.55e-05
		no quantization	7.77e-04	8.96e-04	8.12e-04	8.10e-04	2.07e-05	2.82e-05
	Flash_SD3	int8	1.17e-04	1.70e-04	1.26e-04	1.23e-04	1.10e-05	4.61e-06
		no quantization	1.17e-04	1.70e-04	1.27e-04	1.24e-04	9.83e-06	5.87e-06
	Lumina	int8	4.02e-03	4.23e-03	4.08e-03	4.08e-03	3.83e-05	4.85e-05
		no quantization	4.00e-03	4.19e-03	4.08e-03	4.08e-03	3.92e-05	4.99e-05
	Flux_1	int8	6.20e-04	7.63e-04	6.79e-04	6.77e-04	3.33e-05	4.20e-05
		no quantization	5.74e-04	9.45e-04	7.28e-04	7.18e-04	8.73e-05	7.76e-05

Table G.13 reports the percentage variation in energy consumption of diffusion models when moving from 512×512 to 1024×1024 resolution. As expected, energy consumption increases with higher resolution. This trend is observed across all models, although the magnitude of the increase varies depending on the model. For some models, such as Lumina or SD_1.5, the increase is relatively large, whereas for others, like the PixArt family, it is comparatively small or nearly negligible.

See **Tables G.11** and **G.12**.

Fig. G.17 illustrates the additional experiments conducted on PixArt_Alpha, PixArt_Sigma, and Flash_PixArt models to further investigate the impact of image resolution on energy consumption. We can notice that for all models the energy consumption for non-square resolution is much higher compared to square resolutions.

Appendix H. Prompt length

Before performing the correlation analysis between the length of prompts and the energy consumed during the image generation process, we verified if the energy consumption for each model followed or not a normal distribution, by examining Quantile–Quantile (QQ) plots, which are illustrated in **Fig. H.18**. If the data closely follows the normal distribution, the points on the Q-Q plot will move on a diagonal line (depicted in red), while deviations from the reference line indicate departures from the expected distribution. **Fig. H.18** shows that for all the analyzed models, data does not follow a normal distribution, as the points deviate significantly from the straight diagonal line.

This step ensured the validity of applying Friedman test, which is a non-parametric statistical test, which does not assume a normally distributed data.

Moreover, **Fig. H.19** show the box plots which represent the distribution of energy consumption across different models for the three different prompt lengths. As we can notice, the median energy consumption remains stable across different prompt lengths for all models, with no clear increasing or decreasing trend. This observation suggests that prompt length does not have a significant influence on energy consumption, as no evident pattern is observed across different models. This finding is also supported by the Friedman statistical test. Specifically, the Friedman test is a non-parametric statistical method used to assess whether significant differences exist among multiple treatments or conditions measured on the same or matched subjects. In this case, the test results are appropriate because the short, medium, and long versions are derived from the same set of 30 original prompts. Therefore, the three prompt-length conditions are not independent, but rather represent repeated measurements associated with the same underlying prompts, which can be jointly evaluated using the Friedman test. **Table H.14** presents the result of the test, which examines whether variations in prompt lengths lead to statistically significant differences in energy consumption. Overall, the results confirm that prompt length does not have a statistically significant effect on energy consumption for the vast majority of the considered models. Before multiple-comparison correction, statistically significant differences are observed only for Flux_1 (p-value=0.024) and SDXL_Turbo (p-value=0.045). However, since the Friedman test is performed independently for 17 models, we

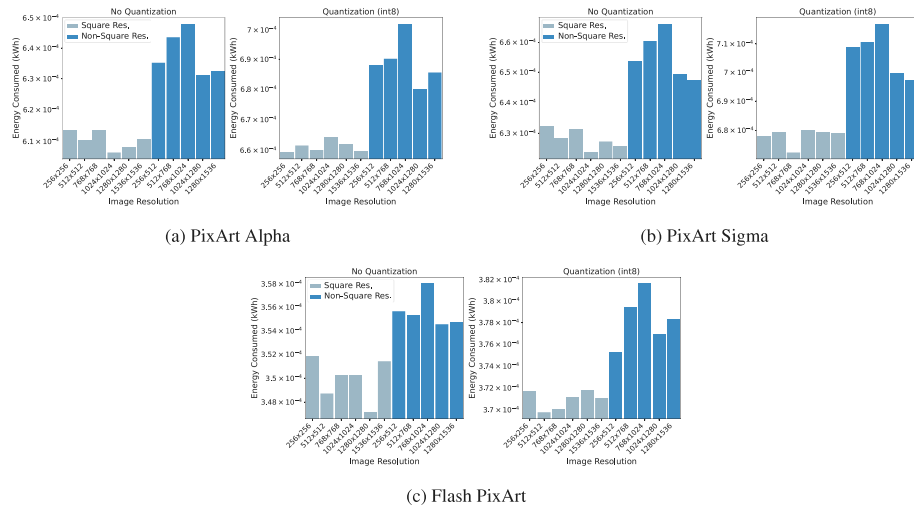


Fig. G.17. Energy consumption (kWh) of PixArt_Alpha, PixArt_Sigma, and Flash_PixArt models at varying image resolution. Bar charts (log scale) show the energy consumed across different image resolutions, categorized as square and non-square resolutions, at fixed quantization setting. Each bar shows the median over 30 prompts.

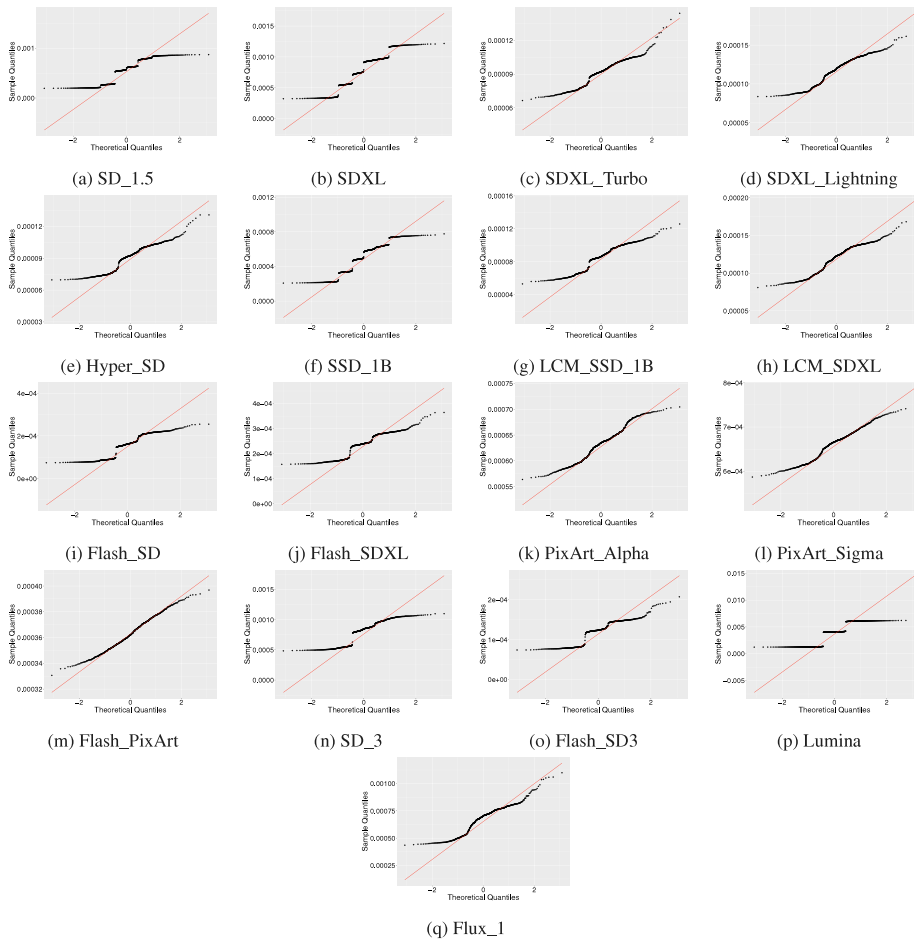


Fig. H.18. Q-Q plots comparing the energy consumption of the analyzed models across various prompt length to the normal distribution. Each plot visualizes the distribution of energy consumption values for a specific model, assessing how closely they follow a normal distribution.

Table F.10

Energy consumption statistics (in kWh) for each model at **fixed 1024×1024 resolution**, comparing INT8 quantization and no quantization runs. Reported values include minimum, maximum, mean, median, standard deviation, and interquartile range (IQR), providing a detailed view of variability across prompts.

Type	Model	Quantization	Min	Max	Mean	Median	Std	IQR
U-Net-based	SD_1.5	int8	8.32e-04	8.73e-04	8.52e-04	8.52e-04	9.20e-06	1.24e-05
		no quantization	7.25e-04	8.29e-04	7.85e-04	7.86e-04	1.52e-05	1.99e-05
	SDXL	int8	1.16e-03	1.22e-03	1.18e-03	1.18e-03	1.27e-05	1.80e-05
		no quantization	9.44e-04	1.01e-03	9.69e-04	9.68e-04	1.24e-05	1.89e-05
	SDXL_Turbo	int8	9.51e-05	1.23e-04	1.01e-04	1.01e-04	4.82e-06	4.44e-06
		no quantization	9.84e-05	1.44e-04	1.05e-04	1.04e-04	7.04e-06	3.74e-06
	SDXL_Lightning	int8	1.31e-04	1.62e-04	1.40e-04	1.39e-04	5.37e-06	5.76e-06
		no quantization	1.25e-04	1.61e-04	1.32e-04	1.31e-04	6.18e-06	5.32e-06
	Hyper_SD	int8	9.61e-05	1.20e-04	1.02e-04	1.01e-04	4.04e-06	2.83e-06
		no quantization	9.79e-05	1.31e-04	1.05e-04	1.04e-04	5.71e-06	4.12e-06
	SSD_1B	int8	7.29e-04	7.77e-04	7.46e-04	7.45e-04	8.64e-06	1.17e-05
		no quantization	6.06e-04	6.53e-04	6.33e-04	6.32e-04	1.05e-05	1.61e-05
	LCM_SSD_1B	int8	9.79e-05	1.20e-04	1.05e-04	1.04e-04	4.35e-06	3.84e-06
		no quantization	9.15e-05	1.26e-04	9.92e-05	9.82e-05	5.23e-06	3.71e-06
	LCM_SDXL	int8	1.32e-04	1.61e-04	1.41e-04	1.40e-04	5.23e-06	4.21e-06
		no quantization	1.27e-04	1.68e-04	1.36e-04	1.35e-04	7.01e-06	4.67e-06
	Flash_SD	int8	1.90e-04	2.29e-04	2.10e-04	2.10e-04	6.74e-06	9.36e-06
		no quantization	2.09e-04	2.55e-04	2.24e-04	2.21e-04	1.05e-05	9.17e-06
	Flash_SDXL	int8	2.74e-04	3.16e-04	2.87e-04	2.85e-04	8.01e-06	8.75e-06
		no quantization	2.65e-04	3.64e-04	2.87e-04	2.80e-04	2.28e-05	1.33e-05
Transformer-based	PixArt_Alpha	int8	6.27e-04	6.74e-04	6.45e-04	6.43e-04	1.04e-05	1.28e-05
		no quantization	5.64e-04	6.21e-04	5.94e-04	5.96e-04	1.28e-05	1.63e-05
	PixArt_Sigma	int8	6.50e-04	7.03e-04	6.77e-04	6.76e-04	1.29e-05	1.85e-05
		no quantization	5.87e-04	6.46e-04	6.17e-04	6.16e-04	1.25e-05	1.76e-05
	Flash_PixArt	int8	3.57e-04	3.89e-04	3.71e-04	3.71e-04	7.67e-06	9.53e-06
		no quantization	3.36e-04	3.65e-04	3.50e-04	3.50e-04	6.60e-06	8.69e-06
	SD_3	int8	9.84e-04	1.10e-03	1.05e-03	1.05e-03	2.26e-05	2.69e-05
		no quantization	9.33e-04	1.03e-03	9.75e-04	9.74e-04	2.54e-05	3.45e-05
	Flash_SD3	int8	1.42e-04	2.08e-04	1.50e-04	1.47e-04	1.17e-05	3.95e-06
		no quantization	1.40e-04	1.92e-04	1.51e-04	1.48e-04	1.07e-05	5.02e-06
	Lumina	int8	5.98e-03	6.21e-03	6.10e-03	6.10e-03	5.22e-05	7.04e-05
		no quantization	6.01e-03	6.20e-03	6.09e-03	6.09e-03	4.36e-05	5.84e-05
	Flux_1	int8	7.42e-04	8.31e-04	7.79e-04	7.74e-04	2.07e-05	3.29e-05
		no quantization	6.93e-04	1.10e-03	8.14e-04	7.96e-04	9.66e-05	8.42e-05

Table G.11

Energy consumption (in kWh) statistics for each model at **fixed int8 quantization**, comparing 512×512 , 768×1024 , and 1024×1024 runs. Reported values include minimum, maximum, mean, median, standard deviation, and interquartile range (IQR), providing a detailed view of variability across prompts.

Type	Model	Resolution	Min	Max	Mean	Median	Std	IQR
U-Net-based	SD_1.5	512×512	2.54e-04	2.89e-04	2.72e-04	2.72e-04	6.89e-06	7.43e-06
		768×1024	6.05e-04	6.42e-04	6.25e-04	6.26e-04	8.28e-06	1.24e-05
		1024×1024	8.32e-04	8.73e-04	8.52e-04	8.52e-04	9.20e-06	1.24e-05
	SDXL	512×512	5.32e-04	5.78e-04	5.54e-04	5.54e-04	9.55e-06	1.52e-05
		768×1024	9.08e-04	9.65e-04	9.34e-04	9.32e-04	1.13e-05	1.52e-05
		1024×1024	1.16e-03	1.22e-03	1.18e-03	1.18e-03	1.27e-05	1.80e-05
	SDXL_Turbo	512×512	6.65e-05	9.66e-05	7.48e-05	7.45e-05	5.10e-06	3.77e-06
		768×1024	8.64e-05	1.17e-04	9.19e-05	9.11e-05	5.28e-06	3.76e-06
		1024×1024	9.51e-05	1.23e-04	1.01e-04	1.01e-04	4.82e-06	4.44e-06
	SDXL_Lightning	512×512	9.28e-05	1.16e-04	1.00e-04	9.91e-05	4.80e-06	5.49e-06
		768×1024	1.15e-04	1.45e-04	1.24e-04	1.24e-04	5.33e-06	4.21e-06
		1024×1024	1.31e-04	1.62e-04	1.40e-04	1.39e-04	5.37e-06	5.76e-06
	Hyper_SD	512×512	6.96e-05	8.68e-05	7.44e-05	7.42e-05	3.34e-06	2.49e-06
		768×1024	8.58e-05	1.06e-04	9.15e-05	9.06e-05	3.91e-06	3.53e-06
		1024×1024	9.61e-05	1.20e-04	1.02e-04	1.01e-04	4.04e-06	2.83e-06
	SSD_1B	512×512	3.22e-04	3.71e-04	3.38e-04	3.37e-04	7.84e-06	7.54e-06
		768×1024	5.61e-04	6.17e-04	5.88e-04	5.90e-04	1.01e-05	1.41e-05
		1024×1024	7.29e-04	7.77e-04	7.46e-04	7.45e-04	8.64e-06	1.17e-05
	LCM_SSD_1B	512×512	6.32e-05	8.49e-05	6.83e-05	6.74e-05	3.92e-06	2.60e-06
		768×1024	8.26e-05	1.12e-04	9.02e-05	8.95e-05	4.42e-06	4.06e-06
		1024×1024	9.79e-05	1.20e-04	1.05e-04	1.04e-04	4.35e-06	3.84e-06
	LCM_SDXL	512×512	9.44e-05	1.16e-04	1.01e-04	1.01e-04	4.14e-06	5.16e-06
		768×1024	1.18e-04	1.48e-04	1.27e-04	1.26e-04	5.82e-06	5.12e-06
		1024×1024	1.32e-04	1.61e-04	1.41e-04	1.40e-04	5.23e-06	4.21e-06
	Flash_SD	512×512	8.49e-05	9.61e-05	8.97e-05	8.94e-05	2.69e-06	3.68e-06
		768×1024	1.46e-04	1.72e-04	1.60e-04	1.61e-04	7.18e-06	1.15e-05
		1024×1024	1.90e-04	2.29e-04	2.10e-04	2.10e-04	6.74e-06	9.36e-06
	Flash_SDXL	512×512	1.57e-04	1.84e-04	1.68e-04	1.68e-04	6.86e-06	1.10e-05
		768×1024	2.22e-04	2.52e-04	2.36e-04	2.35e-04	6.19e-06	7.06e-06
		1024×1024	2.74e-04	3.16e-04	2.87e-04	2.85e-04	8.01e-06	8.75e-06
Transformer-based	PixArt_Alpha	512×512	6.23e-04	6.70e-04	6.43e-04	6.42e-04	1.12e-05	1.56e-05
		768×1024	6.63e-04	7.04e-04	6.85e-04	6.86e-04	8.58e-06	1.20e-05
		1024×1024	6.27e-04	6.74e-04	6.45e-04	6.43e-04	1.04e-05	1.28e-05
	PixArt_Sigma	512×512	6.52e-04	7.05e-04	6.75e-04	6.75e-04	1.13e-05	1.58e-05
		768×1024	6.85e-04	7.43e-04	7.12e-04	7.10e-04	1.37e-05	2.13e-05
		1024×1024	6.50e-04	7.03e-04	6.77e-04	6.76e-04	1.29e-05	1.85e-05
	Flash_PixArt	512×512	3.56e-04	3.84e-04	3.69e-04	3.68e-04	6.74e-06	9.50e-06
		768×1024	3.65e-04	3.97e-04	3.80e-04	3.80e-04	7.55e-06	1.02e-05
		1024×1024	3.57e-04	3.89e-04	3.71e-04	3.71e-04	7.67e-06	9.53e-06
	SD_3	512×512	5.19e-04	5.89e-04	5.49e-04	5.48e-04	1.44e-05	2.14e-05
		768×1024	8.34e-04	9.20e-04	8.71e-04	8.68e-04	1.74e-05	2.55e-05
		1024×1024	9.84e-04	1.10e-03	1.05e-03	1.05e-03	2.26e-05	2.69e-05
	Flash_SD3	512×512	7.43e-05	1.23e-04	8.28e-05	7.94e-05	1.04e-05	3.92e-06
		768×1024	1.17e-04	1.70e-04	1.26e-04	1.23e-04	1.10e-05	4.61e-06
		1024×1024	1.42e-04	2.08e-04	1.50e-04	1.47e-04	1.17e-05	3.95e-06
	Lumina	512×512	1.23e-03	1.37e-03	1.29e-03	1.28e-03	2.99e-05	3.31e-05
		768×1024	4.02e-03	4.23e-03	4.08e-03	4.08e-03	3.83e-05	4.85e-05
		1024×1024	5.98e-03	6.21e-03	6.10e-03	6.10e-03	5.22e-05	7.04e-05
	Flux_1	512×512	4.35e-04	5.37e-04	4.79e-04	4.74e-04	2.54e-05	4.10e-05
		768×1024	6.20e-04	7.63e-04	6.79e-04	6.77e-04	3.33e-05	4.20e-05
		1024×1024	7.42e-04	8.31e-04	7.79e-04	7.74e-04	2.07e-05	3.29e-05

Table G.12

Energy consumption (in kWh) statistics for each model at **fixed no quantization**, comparing 512×512 , 768×1024 , and 1024×1024 runs. Reported values include minimum, maximum, mean, median, standard deviation, and interquartile range (IQR), providing a detailed view of variability across prompts.

Type	Model	Resolution	Min	Max	Mean	Median	Std	IQR
U-Net-based	SD_1.5	512×512	1.93e-04	2.15e-04	2.04e-04	2.04e-04	4.79e-06	6.09e-06
		768×1024	5.25e-04	5.76e-04	5.50e-04	5.51e-04	1.04e-05	1.40e-05
		1024×1024	7.25e-04	8.29e-04	7.85e-04	7.86e-04	1.52e-05	1.99e-05
	SDXL	512×512	3.23e-04	3.84e-04	3.38e-04	3.36e-04	1.02e-05	1.08e-05
		768×1024	7.04e-04	7.93e-04	7.36e-04	7.36e-04	1.46e-05	1.84e-05
		1024×1024	9.44e-04	1.01e-03	9.69e-04	9.68e-04	1.24e-05	1.89e-05
	SDXL_Turbo	512×512	7.08e-05	1.16e-04	7.92e-05	7.84e-05	7.58e-06	4.24e-06
		768×1024	8.69e-05	1.32e-04	9.48e-05	9.30e-05	7.76e-06	4.05e-06
		1024×1024	9.84e-05	1.44e-04	1.05e-04	1.04e-04	7.04e-06	3.74e-06
	SDXL_Lightning	512×512	8.37e-05	1.27e-04	8.96e-05	8.87e-05	6.77e-06	2.96e-06
		768×1024	1.09e-04	1.51e-04	1.16e-04	1.15e-04	6.77e-06	4.45e-06
		1024×1024	1.25e-04	1.61e-04	1.32e-04	1.31e-04	6.18e-06	5.32e-06
	Hyper_SD	512×512	7.01e-05	1.04e-04	7.63e-05	7.53e-05	6.08e-06	5.17e-06
		768×1024	8.78e-05	1.25e-04	9.50e-05	9.38e-05	6.25e-06	3.14e-06
		1024×1024	9.79e-05	1.31e-04	1.05e-04	1.04e-04	5.71e-06	4.12e-06
	SSD_1B	512×512	2.08e-04	2.49e-04	2.19e-04	2.18e-04	6.40e-06	8.15e-06
		768×1024	4.58e-04	5.10e-04	4.83e-04	4.82e-04	8.77e-06	9.49e-06
		1024×1024	6.06e-04	6.53e-04	6.33e-04	6.32e-04	1.05e-05	1.61e-05
	LCM_SSD_1B	512×512	5.30e-05	8.81e-05	6.13e-05	6.05e-05	5.66e-06	3.31e-06
		768×1024	7.96e-05	1.09e-04	8.48e-05	8.41e-05	4.79e-06	2.12e-06
		1024×1024	9.15e-05	1.26e-04	9.92e-05	9.82e-05	5.23e-06	3.71e-06
	LCM_SDXL	512×512	8.09e-05	1.25e-04	9.06e-05	8.94e-05	7.23e-06	4.24e-06
		768×1024	1.07e-04	1.56e-04	1.18e-04	1.16e-04	7.79e-06	5.52e-06
		1024×1024	1.27e-04	1.68e-04	1.36e-04	1.35e-04	7.01e-06	4.67e-06
	Flash_SD	512×512	7.41e-05	1.18e-04	8.09e-05	7.88e-05	8.02e-06	3.19e-06
		768×1024	1.48e-04	2.10e-04	1.65e-04	1.63e-04	1.30e-05	1.85e-05
		1024×1024	2.09e-04	2.55e-04	2.24e-04	2.21e-04	1.05e-05	9.17e-06
	Flash_SDXL	512×512	1.59e-04	2.68e-04	1.82e-04	1.74e-04	2.26e-05	1.07e-05
		768×1024	2.27e-04	3.13e-04	2.46e-04	2.41e-04	1.85e-05	7.49e-06
		1024×1024	2.65e-04	3.64e-04	2.87e-04	2.80e-04	2.28e-05	1.33e-05
Transformer-based	PixArt_Alpha	512×512	5.68e-04	6.19e-04	5.93e-04	5.93e-04	1.17e-05	1.73e-05
		768×1024	6.06e-04	6.59e-04	6.32e-04	6.31e-04	1.12e-05	1.47e-05
		1024×1024	5.64e-04	6.21e-04	5.94e-04	5.96e-04	1.28e-05	1.63e-05
	PixArt_Sigma	512×512	5.90e-04	6.54e-04	6.20e-04	6.20e-04	1.37e-05	1.87e-05
		768×1024	6.39e-04	6.89e-04	6.62e-04	6.63e-04	1.02e-05	1.67e-05
		1024×1024	5.87e-04	6.46e-04	6.17e-04	6.16e-04	1.25e-05	1.76e-05
	Flash_PixArt	512×512	3.31e-04	3.68e-04	3.50e-04	3.50e-04	7.57e-06	1.04e-05
		768×1024	3.41e-04	3.75e-04	3.58e-04	3.57e-04	6.64e-06	8.60e-06
		1024×1024	3.36e-04	3.65e-04	3.50e-04	3.50e-04	6.60e-06	8.69e-06
	SD_3	512×512	4.83e-04	6.28e-04	5.10e-04	5.06e-04	2.09e-05	1.55e-05
		768×1024	7.77e-04	8.96e-04	8.12e-04	8.10e-04	2.07e-05	2.82e-05
		1024×1024	9.33e-04	1.03e-03	9.75e-04	9.74e-04	2.54e-05	3.45e-05
	Flash_SD3	512×512	7.41e-05	1.26e-04	8.29e-05	8.03e-05	1.08e-05	3.61e-06
		768×1024	1.17e-04	1.70e-04	1.27e-04	1.24e-04	9.83e-06	5.87e-06
		1024×1024	1.40e-04	1.92e-04	1.51e-04	1.48e-04	1.07e-05	5.02e-06
	Lumina	512×512	1.24e-03	1.36e-03	1.28e-03	1.28e-03	2.68e-05	3.29e-05
		768×1024	4.00e-03	4.19e-03	4.08e-03	4.08e-03	3.92e-05	4.99e-05
		1024×1024	6.01e-03	6.20e-03	6.09e-03	6.09e-03	4.36e-05	5.84e-05
	Flux_1	512×512	4.55e-04	8.51e-04	5.65e-04	5.31e-04	9.24e-05	9.68e-05
		768×1024	5.74e-04	9.45e-04	7.28e-04	7.18e-04	8.73e-05	7.76e-05
		1024×1024	6.93e-04	1.10e-03	8.14e-04	7.96e-04	9.66e-05	8.42e-05

Table G.13

Percentage variation of energy consumption of diffusion models at varying resolution levels. Variation is computed as the difference between 1024×1024 and 512×512 resolution for each quantization level. Negative values indicate lower energy consumption at 1024×1024 resolution compared with 512×512 . The reported 95% confidence intervals quantify the uncertainty associated with the estimated percentage variations, with narrower intervals indicating more precise estimates.

Type	Model	int8	no quantization		
		Variation (%)	95% CI	Variation (%)	95% CI
U-Net-based	SD_1.5	212.98	[211.64,214.51]	284.87	[281.88,288.42]
	SDXL	113.84	[112.51,115.19]	187.67	[185.93,189.42]
	SDXL_Turbo	35.29	[33.88,37.24]	32.82	[31.11,34.27]
	SDXL_Lightning	40.57	[38.21,41.91]	48.25	[46.66,49.78]
	Hyper_SD	36.71	[35.32,38.38]	38.40	[35.95,40.86]
	SSD_1B	121.22	[120.05,122.80]	189.45	[186.63,193.02]
	LCM_SSD_1B	53.91	[52.43,55.57]	62.51	[60.24,65.31]
	LCM_SDXL	38.52	[36.06,40.66]	50.74	[48.88,52.06]
	Flash_SD	134.69	[132.04,137.85]	179.88	[177.25,182.62]
	Flash_SDXL	69.73	[68.78,71.77]	60.40	[58.71,62.11]
Transformer-based	PixArt_Alpha	0.10	[−0.70,0.99]	0.44	[−0.21,1.09]
	PixArt_Sigma	0.17	[−0.57,1.01]	−0.58	[−1.18,0.34]
	Flash_PixArt	0.71	[−0.18,1.50]	−0.07	[−0.91,0.87]
	SD_3	91.52	[89.22,93.20]	92.51	[90.53,94.06]
	Flash_SD3	85.02	[82.65,86.63]	84.50	[82.82,86.28]
	Lumina	375.52	[373.20,378.23]	375.12	[373.27,377.88]
	Flux_1	63.23	[60.42,66.51]	49.18	[44.79,52.70]

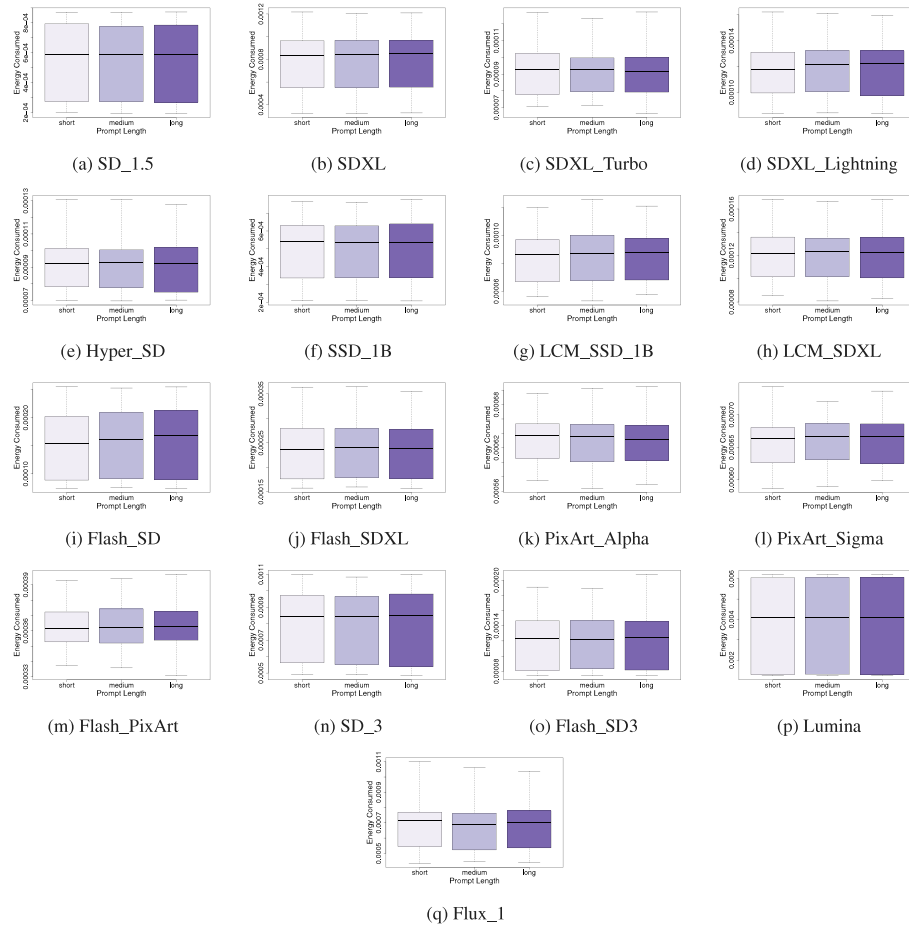
**Fig. H.19.** Box plots of the energy consumption across various prompt length for the analyzed models.

Table H.14

Friedman test results assessing the effect of prompt length (short, medium, and long) on energy consumption. χ^2 denotes the Friedman test statistic, Kendall's W measures the effect size, and the reported p -values correspond to the uncorrected and Holm-corrected tests.

Model	χ^2	Kendall's W	P-value	P-value (Holm corr.)
Flash_PixArt	0.200	0.003	0.905	1.000
Flash_SD	3.800	0.063	0.150	1.000
Flash_SD3	0.067	0.001	0.967	1.000
Flash_SDXL	1.667	0.028	0.435	1.000
Flux_1	7.467	0.124	0.024	0.407
Hyper_SD	0.800	0.013	0.670	1.000
LCM_SDXL	0.800	0.013	0.670	1.000
LCM_SSD_1B	1.267	0.021	0.531	1.000
Lumina	0.600	0.010	0.741	1.000
PixArt_Alpha	4.867	0.081	0.088	1.000
PixArt_Sigma	1.067	0.018	0.587	1.000
SDXL	0.467	0.008	0.792	1.000
SDXL_Lightning	4.200	0.070	0.122	1.000
SDXL_Turbo	6.200	0.103	0.045	0.721
SD_1.5	1.400	0.023	0.497	1.000
SD_3	2.467	0.041	0.291	1.000
SSD_1B	0.467	0.008	0.792	1.000

applied the Holm correction to account for multiple comparisons. After correction, none of the models reaches the conventional significance threshold of $\alpha = 0.05$, with the smallest corrected p -value being 0.407 for Flux_1 and 0.721 for SDXL_Turbo. Furthermore, Kendall's W values are consistently low, with the largest effect size observed for Flux_1 ($W=0.124$), indicating that the practical effect of prompt length is small even where the uncorrected test suggests statistical significance.

Data availability

No data was used for the research described in the article.

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